

Resolution status: all 256² and 256³ simulations are PROVISIONAL — VERY LOW RESOLUTION. Coarser runs are provisional as well. Higher-resolution runs are checks, not evidence of convergence by themselves; physical conclusions remain provisional until supported by resolution comparisons.

NUMERICALWAVETANKS.JL · WC25 WAVE-AVERAGED TURBULENCE · TWO AUTONOMOUS CAMPAIGNS · DELTAAI, 2–3 OCTOBER 2026

Surface stress and the narrow-ring β -plane analogue in WC25 turbulence

Velocity notation: $u'_{\text{rms}}(t) = \sqrt{\langle |\mathbf{u}'_{\text{L}}(t)|^2 \rangle}$ denotes the total rms velocity fluctuation, not one component; $u'_{\text{rms},0}$ is its initial value. The stress ratio is $u^*/u'_{\text{rms},0}$. We reserve q for potential vorticity. Archived figure versions may retain q_0 or q' for these rms speeds; current figures use the updated notation.

Results document for the two October 2026 campaigns that extend Wagner & Constantinou (2025, JFM 1020 A51, "WC25"). Campaign A asks whether a constant along-wave surface stress turns the depth-alternating jets of initially Lagrangian-shear-free wave-averaged turbulence into Langmuir rolls. Campaign B asks whether the two-dimensional (x – z) WC25 dynamics started from the Vallis & Maltrud (1993) spectral ring reproduce the β -plane dumbbell spectrum and zonal organisation of Vallis (2017, Figs 12.3–12.4). Both campaigns ran on DeltaAI GH200 GPUs with Oceananigans 0.107; every figure is Julia/CairoMakie output from the campaign analysis scripts. The two source reports are provided alongside this synthesis.

Summary

A · Zero stress shows WC25-like structures

A counter-wave surface jet ($-3.2\text{e-}3$) and depth-alternating interior bands grow from an initially shear-free Lagrangian field; their share of the relative energy rises from 0.19 ($t = 400$) to 0.40 ($t = 1000$) while the wave-free control stays below 0.02. The generated mean flow is weak relative to the imposed Stokes drift, so $U_E \approx -u^*$ remains a useful approximation; the Lagrangian mean does have jet shear.

A · Stress ratio organises the sampled cases

In the cases run, rolls appeared when the instantaneous ratio $u^*/u'_{\text{rms}}(t)$ reached ≈ 0.7 –1 and only for $La \lesssim 0.3$: $u^*/u'_{\text{rms},0} = 0.01$ stayed indistinguishable from zero stress to $t = 1000$, 0.03 transitioned at $t \approx 277$, 0.1 at $t \approx 80$, and 0.3–1 gave shear turbulence with roll anisotropy ≤ 1.2 , the same as their wave-free controls. These are observations in this sampled setup (one seed at most ratios, two grids at three ratios), not necessary or sufficient conditions.

B · The dumbbell phenomenology is reproduced qualitatively

With shallow waves ($\beta_{\text{eff}} = 1/2$ exactly) the ring energy redistributes anisotropically and piles up near the k_z axis around a depleted region on the k_x axis: anisotropy index $0 \rightarrow 0.69$, centroid $9 \rightarrow 5.2$ cycles at 1024^2 , x -elongated vorticity bands, a measured inverse energy flux. The six-seed $k_0 = 12$, $rb = 2$ ensemble shows the VM93 Fig. 5 dumbbell shape cleanly; this is a source-matched qualitative comparison, not an exact reproduction.

B • Weaker beta gives stronger mean jets in these runs

The exact-zonal energy fraction falls from 0.16 ($rb = 0.5$) to about 0.04 ($rb = 1$) and 0.017 ($rb = 2$) at 100 T_e ; ring width, a second grid (512^2), seed and the WENO regime give the same picture. Deep waves confine β dynamics to the upper third and add a seed-sensitive x-mean jet.

Cost

12.94 billed GPU-hours in total (surface stress 10.83, Vallis-2D 2.11) against a 60 GPU-hour ceiling; this includes validation, retries and analysis.

Methods

Shared numerics

WC25 configuration: wave-averaged (Craig–Leibovich) Navier–Stokes for the Lagrangian-mean velocity u^L in a unit domain, periodic in x (and y in 3-D), impermeable free-slip top ($z = 1$) and bottom, WENO(order 9) implicit LES with no explicit closure, RK3, Float64, Oceananigans 0.107. Stokes drift from the printed WC25 equations: deep $\partial z u^s = \frac{1}{4}e^{8(z-1)}$ ($u^s(1) = 1/32$), shallow/medium $\partial z u^s = z/2$. u^s is never added to the initial condition, so the primary initial Lagrangian mean is shear-free and the Eulerian mean starts at $-u^s$. The repository driver's DeepStokesShear(2, 8) profile is not eq. (2.7) and was not used by either campaign. All JLD2 output uses stream IO (memory-mapped IO fails on Lustre), checkpoints are segmented and were verified to restart bitwise identically.

	A · SURFACE STRESS (3-D)	B · VALLIS RING (2-D, X-Z)
Grid	128 ³ (sweep), 256 ³ (three cases); unit cube	256 ² (all), 512 ² (three cases); Flat y , $v \equiv 0$
Initial condition	Seeded solenoidal random field, spectrum $\propto (k/k_e)^4 e^{-2(k/k_e)^2}$, $k_e = 2\pi \cdot 10$, rms $u'_{rms,0} = 0.03$, zero horizontal means; WC25's 1000 $\rightarrow$ 10 vorticity spin-up skipped by design	Streamfunction ring $\psi = \sum 2\text{Re}[A e^{i2\pi m x}] \sin(n\pi z)$, Gaussian $E(K) \propto e^{-(K-K_0)^2/2\sigma^2}$, $K_0 = 2\pi \cdot 9$ (or 12), $\sigma =$ one mode spacing (a chosen width: VM93's $Z/E = 2K^2$ is consistent with a narrow ring but does not fix σ uniquely), exactly solenoidal on the C-grid, $k_x = 0$ modes removed (they would carry 6% of the energy)
Control parameter	$u^*/u'_{rms,0} \in \{0, 0.01, 0.03, 0.1, 0.3, 1\}$, $\tau = u^{*2}$ as a flux condition on u at $z = 1$; $La = \sqrt{(u^*/u^s(1))}$	$rb = \beta/(u'_{rms,0}K_0^2)$ with $\beta = \partial z z u^s = \frac{1}{2}$ (shallow); $rb = 1 \Rightarrow u'_{rms,0} = 1.56e-4$, turnover $T_e = 1/(u'_{rms,0}K_0) = 113 =$ Rossby time K_0/β
Families	deep waves (primary), medium waves ($\partial z u^s = z/2$), wave-free controls from the same Lagrangian field	shallow (β -plane exact), deep ($\beta(z) = 2e^{8(z-1)}$), surface-intensified), wave-free, plus a doubly periodic pseudo-spectral β -plane reference
Durations	$t = 400$ all; 1000 for deep $r \leq 0.3$, $r = 1$ and the $r = 0$ control	200 T_e pilots; 100 T_e sensitivity; 60 T_e ensemble
Validation	divergence, momentum ($1e-19$), $d\langle U \rangle/dt = \tau/H$, wave-only quiescence, energy monotone, budget residual ≤ 0 , restart	divergence $7e-18$, Parseval-exact spectra, linear Rossby wave within 0.23% with $-x$ phase, quiescence, Stokes work $6e-17$, Δt convergence to $1e-4$ for $\Delta t \leq 4$, restart

Physics of the 2-D analogue

With $\omega_y = \partial z u - \partial x w$, the curl of the wave-averaged momentum equation gives $D(\omega_y - \partial z u^s)/Dt = 0$, so $D\omega_y/Dt = (\partial z z u^s) w$: a β -plane with $z \leftrightarrow y$, $w \leftrightarrow v$, $\zeta = -\omega_y$ and $\beta_{\text{eff}}(z) = \partial z z u^s$. The Stokes momentum terms $\partial z u^s$ ($w \hat{x} - u \hat{z}$) are perpendicular to u^L and do no work on it; energy is conserved up to the implicit dissipation. For shallow waves $\beta_{\text{eff}} = \frac{1}{2}$ is uniform and the wall problem is the odd-symmetric subspace of a doubly periodic 1×2 β -plane; for deep waves β_{eff} varies by 3000 $\times$ over the depth and the case is not a homogeneous- β reproduction.

Case tables

A • Surface stress

U*/U' _{RMS,0}	U*	LA	FAMILY	N	SEEDS	T_END	OUTCOME
0	0	0	(deep, deep · none · medium); undefined (none)	128 ³ , 256 ³ (deep)	1, 2	1000	WC25 regime: counter-wave surface jet, alternating bands, K _{jet} /K _{rel} 0.19 → 0.40
0.01	3e−4	0.10	deep	128 ³	1	1000	indistinguishable from zero stress (u*/u' _{rms} ≤ 0.3)
0.03	9e−4	0.17	deep	128 ³ , 256 ³	1, 2	1000	transition at t ≈ 277 (u*/u' _{rms} = 0.74; 256 ³ : 0.66–0.74), Langmuir state by t = 1000
0.1	3e−3	0.31	deep · none · medium	128 ³ , 256 ³ (deep)	1	400 (1000 deep)	rolls from t ≈ 80 (u*/u' _{rms} = 1.0; 256 ³ onset t = 156, ratio 1.06)
0.3	9e−3	0.54	deep	128 ³	1	1000	shear-dominated, roll anisotropy ≤ 1.2
1	3e−2	0.98	deep · none	128 ³	1, 2	400; deep extension to t ≈ 1000 still finishing at report time	shear turbulence, same anisotropy as its control; surface velocity O(1), where the implicit dissipation is velocity-dependent

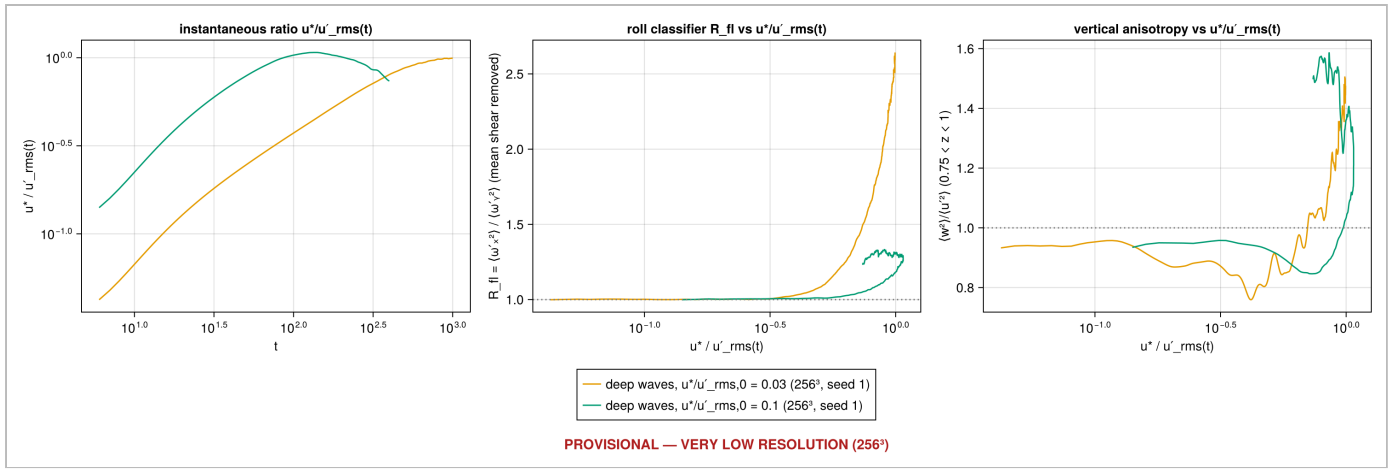
B • Vallis-2D

CASE SET	PARAMETERS	N	SEEDS	T_E	OUTCOME (EXACT-ZONAL K_JET/K · ANISOTROPY AT END)
pilots shallow / deep / none	$k_0 = 9$, $\sigma = 1$, $rb = 1$	256 ² , 512 ²	1	200	0.050 · 0.70 / 0.39 · 0.62 / 0.12 · 0.01 (512 ² : 0.050 · 0.715 / 0.384 / 0.133)
β -plane reference	doubly periodic, $\beta =$ $\frac{1}{2}$, same ring law and $u'_{rms,0}$, ∇^6 hyperviscosity	256 ²	1	200	0.060; same dumbbell-shaped spectrum as the wall case
rb bracket (shallow)	$rb = 0.5$, 2	256 ²	1	100	0.159 · 0.44 and 0.017 · 0.70 (base at 100 T_e $\approx$ 0.04 · 0.62)
ring width (shallow)	$\sigma = 0.5$, 2	256 ²	1	100	0.027 · 0.61 and 0.040 · 0.56
$k_x = 0$ modes kept	+6% initial zonal energy	256 ²	1	100	0.097 · 0.70
$\lambda = 200$ rescaling (shallow, none)	u^s and $u'_{rms,0} \times 200$, $u'_{rms,0} = 0.031$: identical physics, nonlinear WENO regime	256 ²	1	100	0.039 · 0.63 (K/K ₀ 0.990) and 0.118 · 0.01
second seed (shallow, none, deep)	$k_0 = 9$, $rb = 1$	256 ²	2	100	0.043 · 0.64 / 0.113 · 0.01 / 0.074 · 0.45
ensemble (VM93- inspired parameters)	$k_0 = 12$, $\sigma = 1$, $rb = 2$	256 ²	shallow 1–6, none 1–3, deep 1	60	shallow 0.012 ± 0.003; none 0.06 ± 0.03

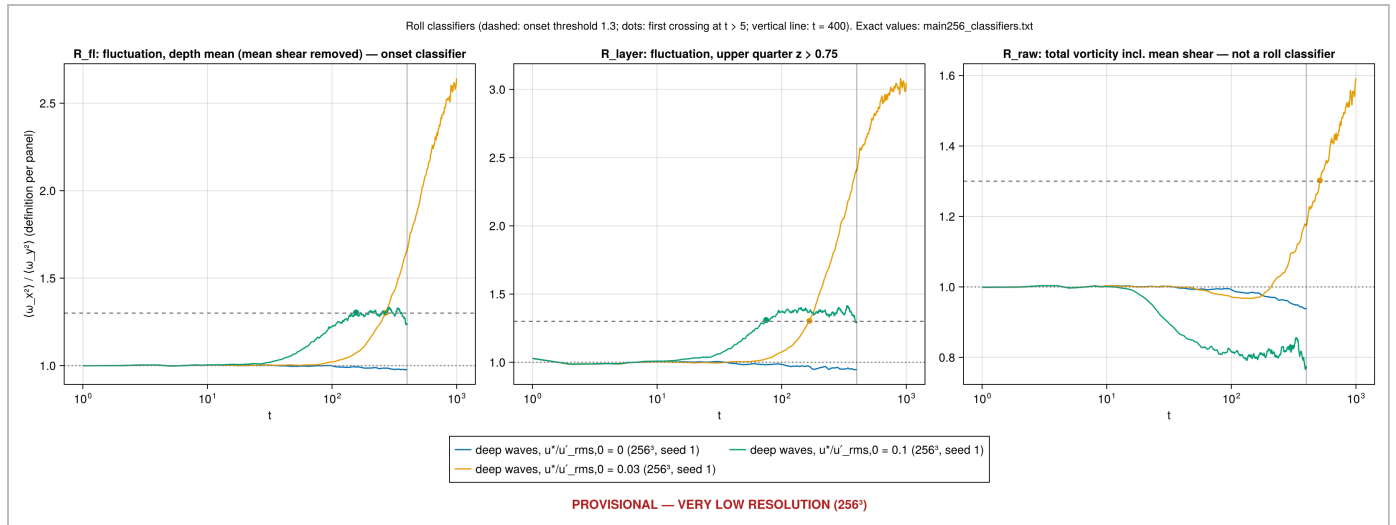
3-D upgrade pending: the 256³ figures directly below are the highest grid completed for these cases (deep waves, $r = 0$, 0.03, 0.1); the 128³ plots further down are retained as provisional legacy results and as the only coverage of the other cases. The 512³ runs at $r = 0$, 0.03 and 0.1 are in progress; the requested regime-diagram upgrade is not complete, and a full 512³ sweep (rough estimate $\approx$ 420 GPU-h) does not fit the 60 GPU-h ceiling, so budget coverage remains incomplete.

A • Surface stress at the highest completed grid (256³)

Deep waves at $u^*/u'_{rms,0} = 0$, 0.03 and 0.1 on 256³. R_{fl} is the fluctuation enstrophy ratio $\langle \omega'^2_x \rangle / \langle \omega'^2_y \rangle$ with the mean-shear contributions $(\partial_z V)^2$, $(\partial_z U)^2$ removed from the horizontal-mean profiles and the depth mean taken over all $N + 1$ faces; it is the onset classifier (threshold 1.3 at $t > 5$). R_{layer} is the same ratio over the upper quarter ($z > 0.75$). $R_{raw} = \langle \omega'^2_x \rangle_V / \langle \omega'^2_y \rangle_V$ uses the total vorticity, including the mean shear, and is not a roll classifier. Exact classifier values are tabulated with the classifier figure; all onset times quoted on this page use R_{fl} . Per-plot resolution audit: REPORT_main_resolution_audit.md.



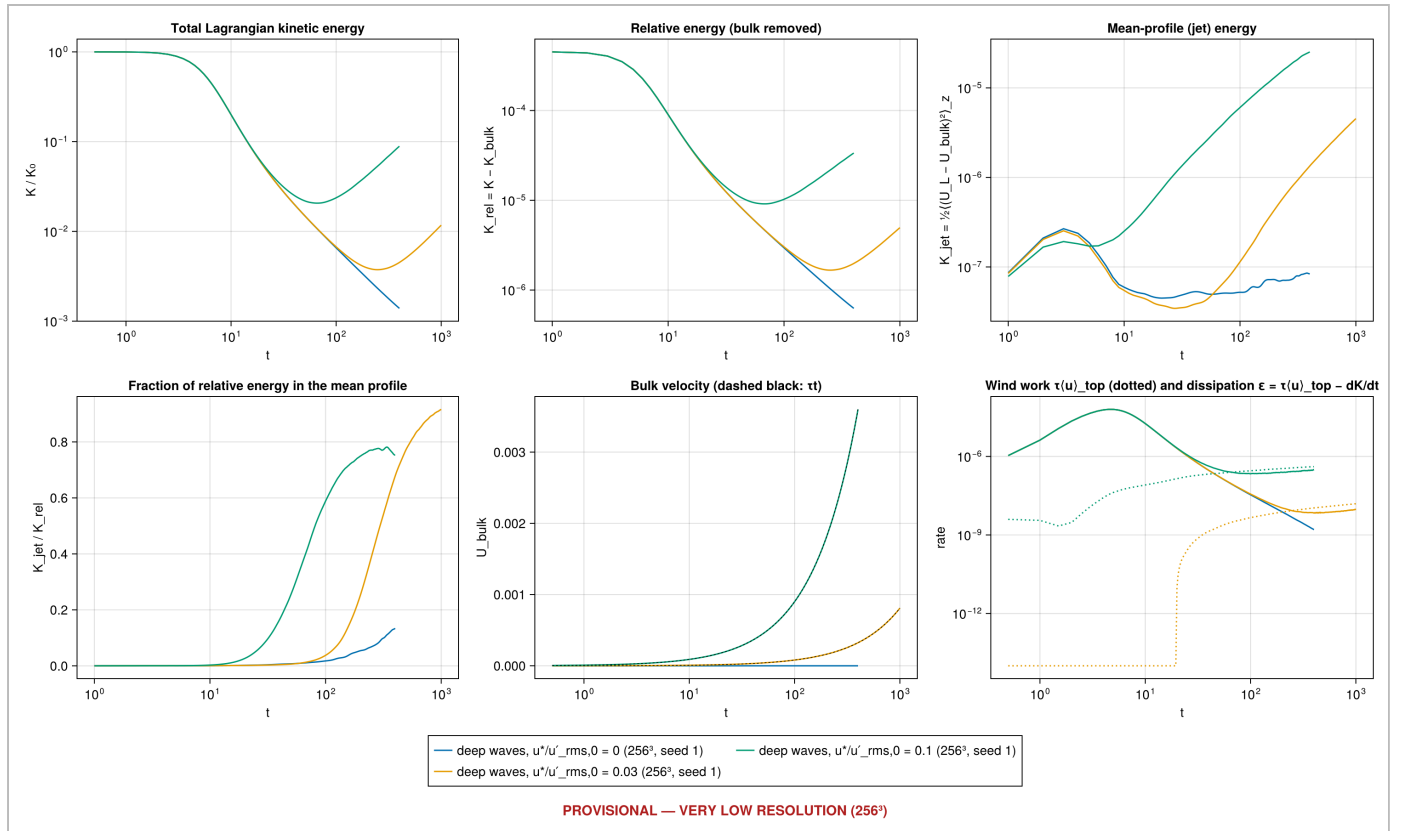
PROVISIONAL — VERY LOW RESOLUTION (256^3). Regime diagram (256^3 , deep waves). $u^*/u'_{rms}(t)$ against time (left), and the roll classifier R_{fl} (middle) and the upper-quarter vertical anisotropy (right) against that ratio, for $r = 0.03$ (to $t = 1000$) and $r = 0.1$ (to $t = 400$); $r = 0$ has no stress ratio and appears in the other figures. R_{fl} first exceeds 1.3 at $t = 270$ for $r = 0.03$ ($u^*/u'_{rms} = 0.66$ there) and at $t = 156$ for $r = 0.1$ (1.06). At $t = 400$, $R_{fl} = 1.658$ ($r = 0.03$, where $u^*/u'_{rms} = 0.80$) and 1.239 ($r = 0.1$); at $r = 0$ it is 0.976. *Provenance:* 256^3 runs `deep_r{0,0p03,0p1}_seed1_N256` (COMPLETED; $r = 0.03$ to $t = 1000$, the others to $t = 400$), all from the 256^3 -draw seed-1 initial field (identical to the legacy 256^3 seed-1 file to 3×10^{-17}). This is a different random realization from the 128^3 legacy sweep below; the matched lower-resolution baseline is the 128^3 run from the same field (REPORT_resolution.md). No wave-free, $r = 0.01, 0.3$ or 1 case exists above 128^3 . Figure main256_regime.png.



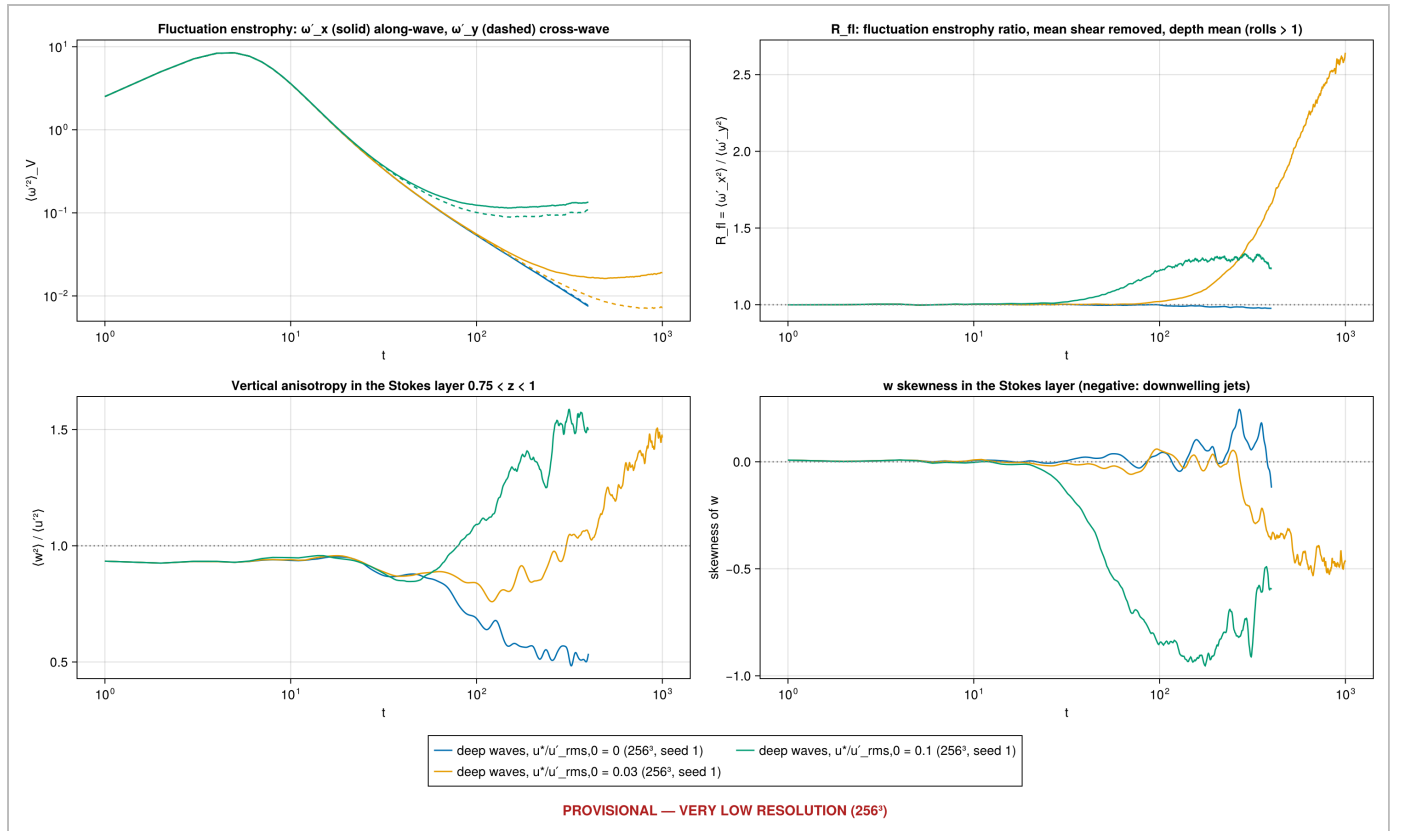
PROVISIONAL — VERY LOW RESOLUTION (256³). Roll classifiers (256³), exact values. R_{fl} (left, the onset classifier), R_{layer} (middle) and R_{raw} (right) against time on logarithmic axes; dashed line: threshold 1.3; dots: first crossing at t > 5; vertical line: t = 400. R_{fl} is the fluctuation enstrophy ratio $\langle \omega_x'^2 \rangle / \langle \omega_y'^2 \rangle$ with the mean-shear contributions $(\partial z V)^2$, $(\partial z U)^2$ removed from the horizontal-mean profiles and the depth mean taken over all N + 1 faces; it is the onset classifier (threshold 1.3 at t > 5). R_{layer} is the same ratio over the upper quarter (z > 0.75). R_{raw} = $\langle \omega_x^2 \rangle_V / \langle \omega_y^2 \rangle_V$ uses the total vorticity, including the mean shear, and is not a roll classifier. Because the mean shear enters the denominator of R_{raw}, R_{raw} stays near or below 1 while rolls are present (r = 0.1: 0.775 at t = 400). Values below are read from the saved profiles and statistics (main256_classifiers.txt), not from the plot.

256 ³ , DEEP	T	R_FL	R_LAYER	R_RAW	ONSET R_FL / R_LAYER / R_RAW
r = 0	400	0.976	0.945	0.937	none / none / none
r = 0.03	200	1.152	1.499	0.993	270 / 167 / 514
	400	1.658	2.404	1.175	
	1000	2.643	3.051	1.592	
r = 0.1	200	1.308	1.378	0.824	156 / 75 / never
	400	1.239	1.296	0.775	

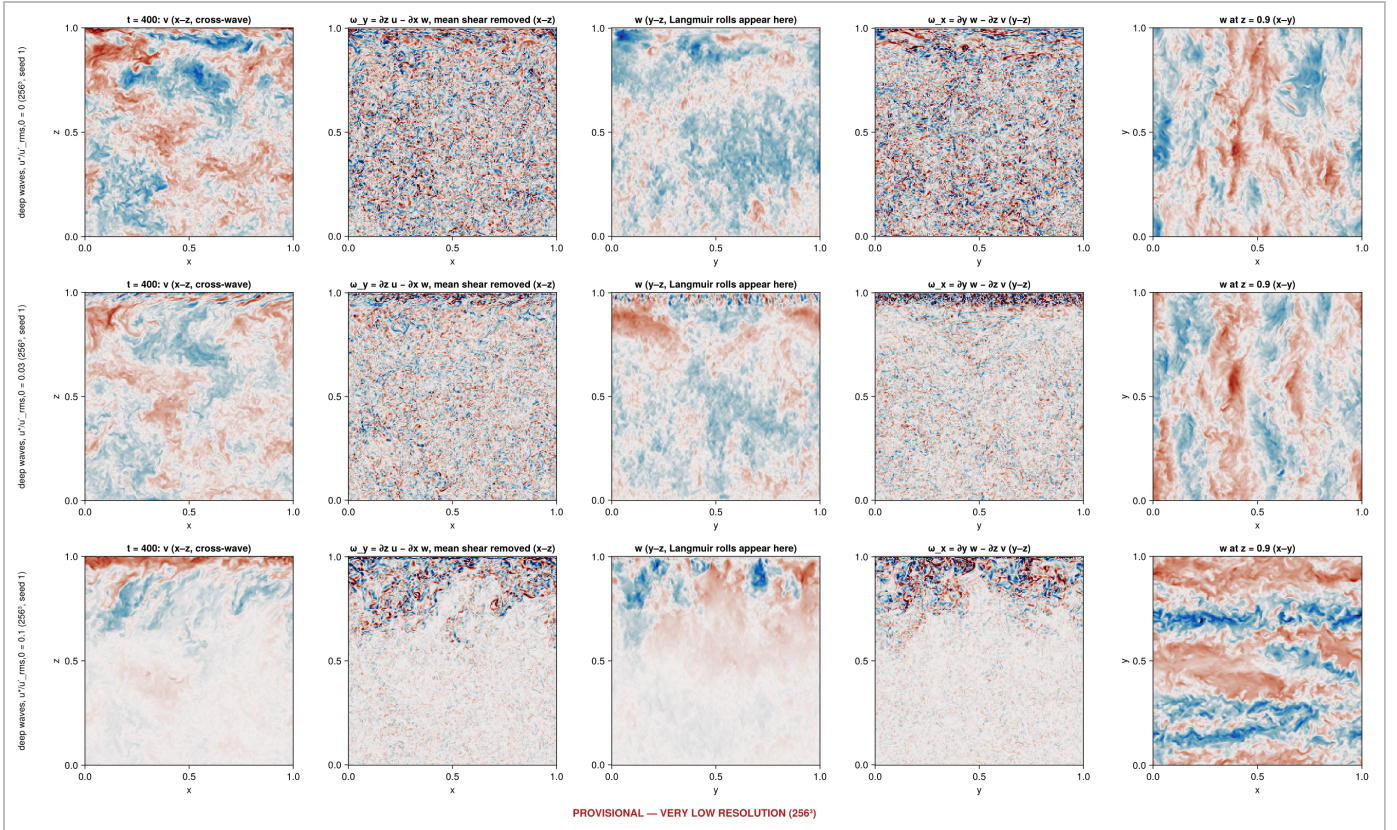
Provenance: 256³ runs deep_r{0,0p03,0p1}_seed1_N256 (COMPLETED; r = 0.03 to t = 1000, the others to t = 400), all from the 256³-draw seed-1 initial field (identical to the legacy 256³ seed-1 file to 3×10⁻¹⁷). This is a different random realization from the 128³ legacy sweep below; the matched lower-resolution baseline is the 128³ run from the same field (REPORT_resolution.md). No wave-free, r = 0.01, 0.3 or 1 case exists above 128³. Figure main256_classifiers.png.



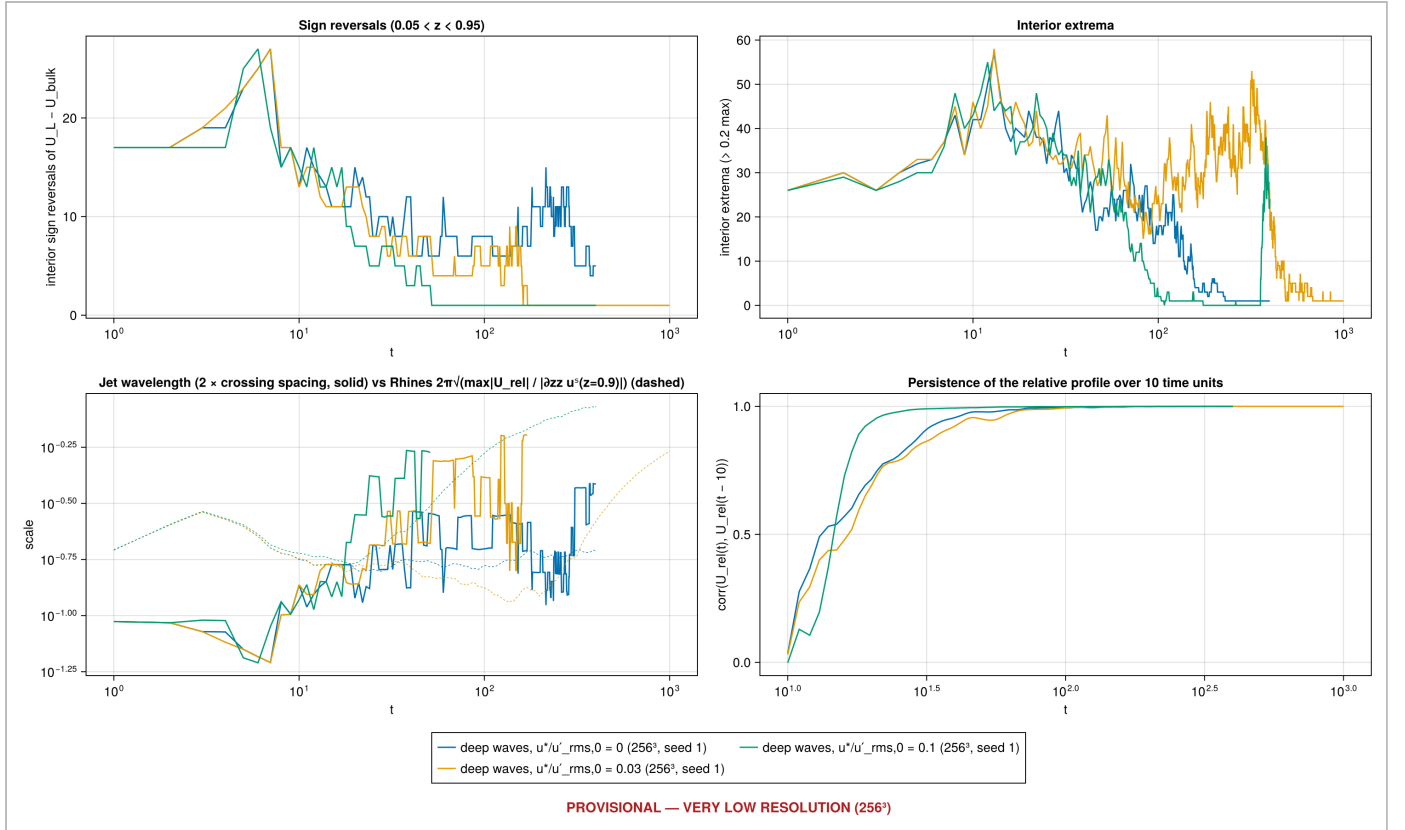
PROVISIONAL — VERY LOW RESOLUTION (256³). Energy budget (256³). Total energy K/K_0 , relative energy, mean-profile (jet) energy, its fraction of the relative energy, the bulk velocity (which follows τt) and the wind work and dissipation. At the end of each run $K/K_0 = 1.39\text{e-}3$ ($r = 0$, $t = 400$), $1.17\text{e-}2$ ($r = 0.03$, $t = 1000$) and $8.88\text{e-}2$ ($r = 0.1$, $t = 400$); $K_{\text{jet}}/K_{\text{rel}} = 0.134$, 0.916 and 0.753 . *Provenance:* 256³ runs `deep_r{0,0p03,0p1}_seed1_N256` (COMPLETED; $r = 0.03$ to $t = 1000$, the others to $t = 400$), all from the 256³-draw seed-1 initial field (identical to the legacy 256³ seed-1 file to 3×10^{-17}). This is a different random realization from the 128³ legacy sweep below; the matched lower-resolution baseline is the 128³ run from the same field (REPORT_resolution.md). No wave-free, $r = 0.01$, 0.3 or 1 case exists above 128³. Figure `main256_energy.png`.



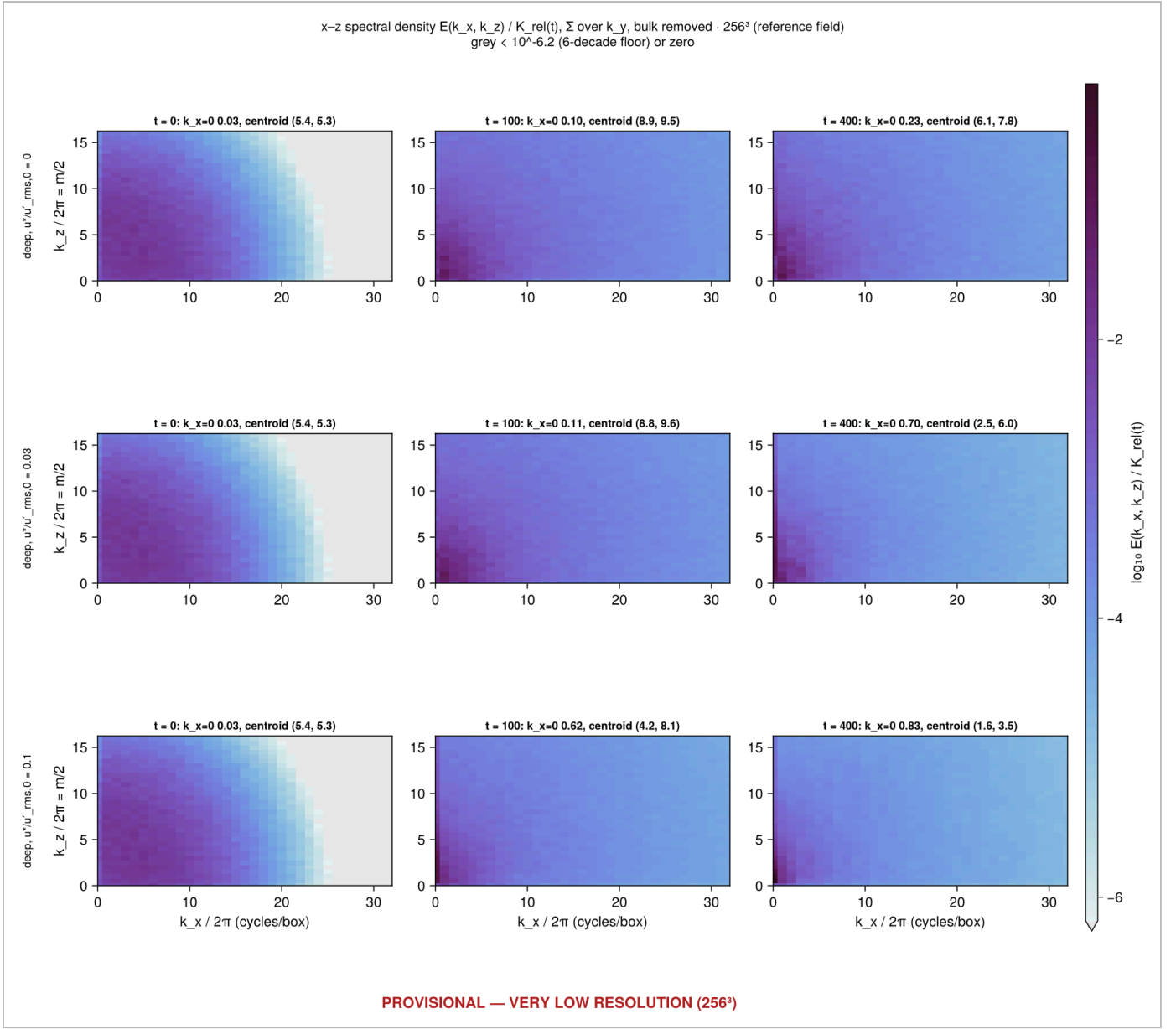
PROVISIONAL — VERY LOW RESOLUTION (256³). Structure diagnostics (256³). Fluctuation enstrophy components (top left), the roll classifier R_{fl} (top right), and the vertical anisotropy and w skewness in the Stokes layer $0.75 < z < 1$ (bottom) against time for $r = 0, 0.03$ and 0.1 . R_{fl} stays between 0.976 and 1.00 for $r = 0$, levels near 1.3 for $r = 0.1$ (1.308 at $t = 200$, 1.239 at $t = 400$), and reaches 1.658 at $t = 400$ and 2.643 at $t = 1000$ for $r = 0.03$ (table above). With stress the Stokes-layer vertical anisotropy rises above 1 and the w skewness becomes negative (downwelling-dominated); these two panels are not tabulated, so no values are quoted. *Provenance:* 256³ runs deep_r{0,0p03,0p1}_seed1_N256 (COMPLETED; $r = 0.03$ to $t = 1000$, the others to $t = 400$), all from the 256³-draw seed-1 initial field (identical to the legacy 256³ seed-1 file to 3×10^{-17}). This is a different random realization from the 128³ legacy sweep below; the matched lower-resolution baseline is the 128³ run from the same field (REPORT_resolution.md). No wave-free, $r = 0.01, 0.3$ or 1 case exists above 128³. Figure main256_anisotropy.png.



PROVISIONAL — VERY LOW RESOLUTION (256³). Sections at $t = 400$ (256³), rows $r = 0, 0.03, 0.1$; columns: cross-wave v in x - z , ω_y with the mean shear removed, w in y - z , ω_x , and w at $z = 0.9$. At $r = 0$ and 0.03 , w at $z = 0.9$ is organised into cross-wave (y -elongated) streaks; at $r = 0.1$ it forms along-wave (x -elongated) bands beneath a vorticity-filled surface layer, the roll signature. Sections at $t = 100$: main256_sections_t100.png. *Provenance:* 256³ runs deep_r{0,0p03,0p1}_seed1_N256 (COMPLETED; $r = 0.03$ to $t = 1000$, the others to $t = 400$), all from the 256³-draw seed-1 initial field (identical to the legacy 256³ seed-1 file to 3×10^{-17}). This is a different random realization from the 128³ legacy sweep below; the matched lower-resolution baseline is the 128³ run from the same field (REPORT_resolution.md). No wave-free, $r = 0.01, 0.3$ or 1 case exists above 128³. Figure main256_sections_t400.png.



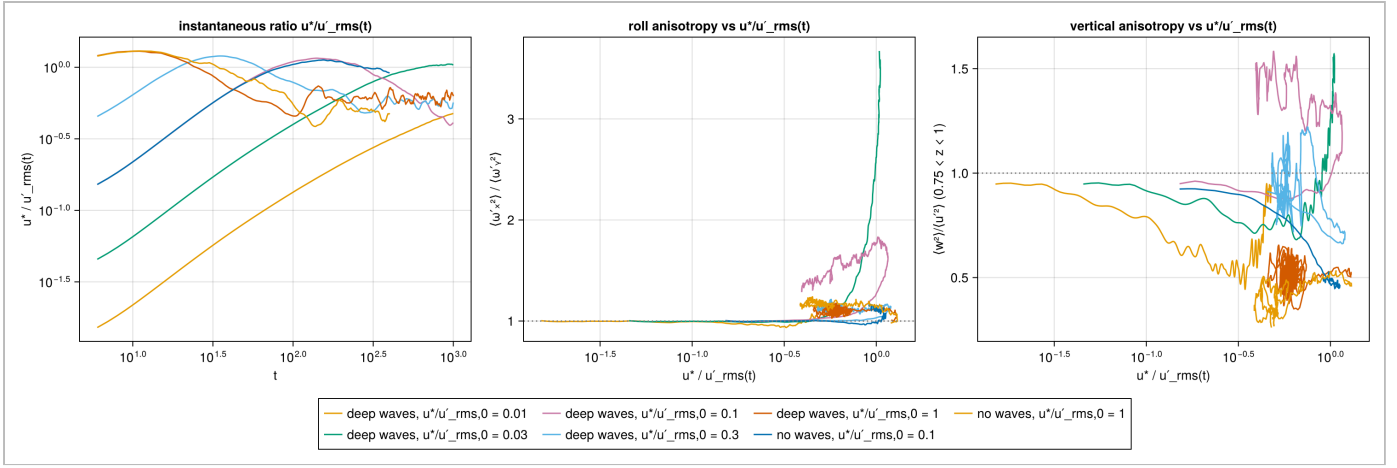
PROVISIONAL — VERY LOW RESOLUTION (256³). Jet metrics (256³). At $r = 0$ the final profile $U_L - U_{\text{bulk}}$ has 5 interior sign reversals with a band scale of 0.385 at $t = 400$ ($\max |U_{\text{rel}}| = 8.4\text{e-}4$); at $r = 0.03$ and 0.1 the wind layer dominates the profile (one reversal). *Provenance:* 256³ runs `deep_r{0,0p03,0p1}_seed1_N256` (COMPLETED; $r = 0.03$ to $t = 1000$, the others to $t = 400$), all from the 256³-draw seed-1 initial field (identical to the legacy 256³ seed-1 file to 3×10^{-17}). This is a different random realization from the 128³ legacy sweep below; the matched lower-resolution baseline is the 128³ run from the same field (REPORT_resolution.md). No wave-free, $r = 0.01, 0.3$ or 1 case exists above 128³. Figure `main256_jets.png`.



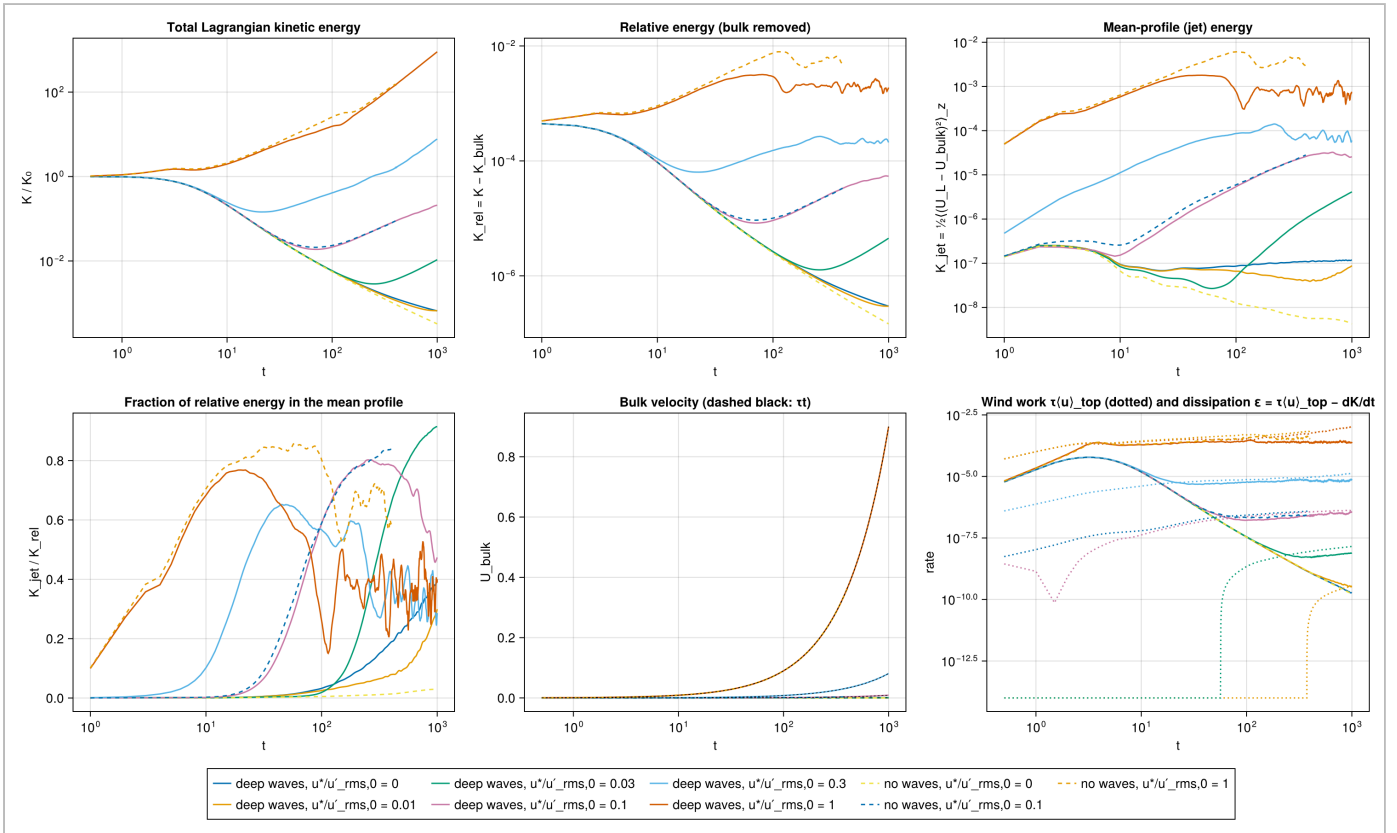
PROVISIONAL — VERY LOW RESOLUTION (256³). Directional spectra in the x-z plane (256³). $E(k_x, k_z)/K_{rel}(t)$ of the velocity fluctuation about the volume mean, summed over k_y (Fourier in x, cosine/sine in z), at $t = 0, 100$ and 400 ; ΣE equals $\frac{1}{2}\langle |u'|^2 \rangle$ to six digits at every time. The $k_x = 0$ fraction at $t = 400$ is 0.230 ($r = 0$), 0.705 ($r = 0.03$) and 0.828 ($r = 0.1$). $t = 1000$ exists at 256³ only for $r = 0.03$ and is not shown. Log₁₀ colour shared by all panels, grey below a six-decade floor, one cell per mode, no overlays. *Provenance:* 256³ runs deep_r{0,0p03,0p1}_seed1_N256 (COMPLETED; $r = 0.03$ to $t = 1000$, the others to $t = 400$), all from the 256³-draw seed-1 initial field (identical to the legacy 256³ seed-1 file to 3×10^{-17}). This is a different random realization from the 128³ legacy sweep below; the matched lower-resolution baseline is the 128³ run from the same field (REPORT_resolution.md). No wave-free, $r = 0.01, 0.3$ or 1 case exists above 128³. Figure main256_xz_spectra_heatmap.png.

A • Legacy comparison — Surface stress on WC25 decaying turbulence

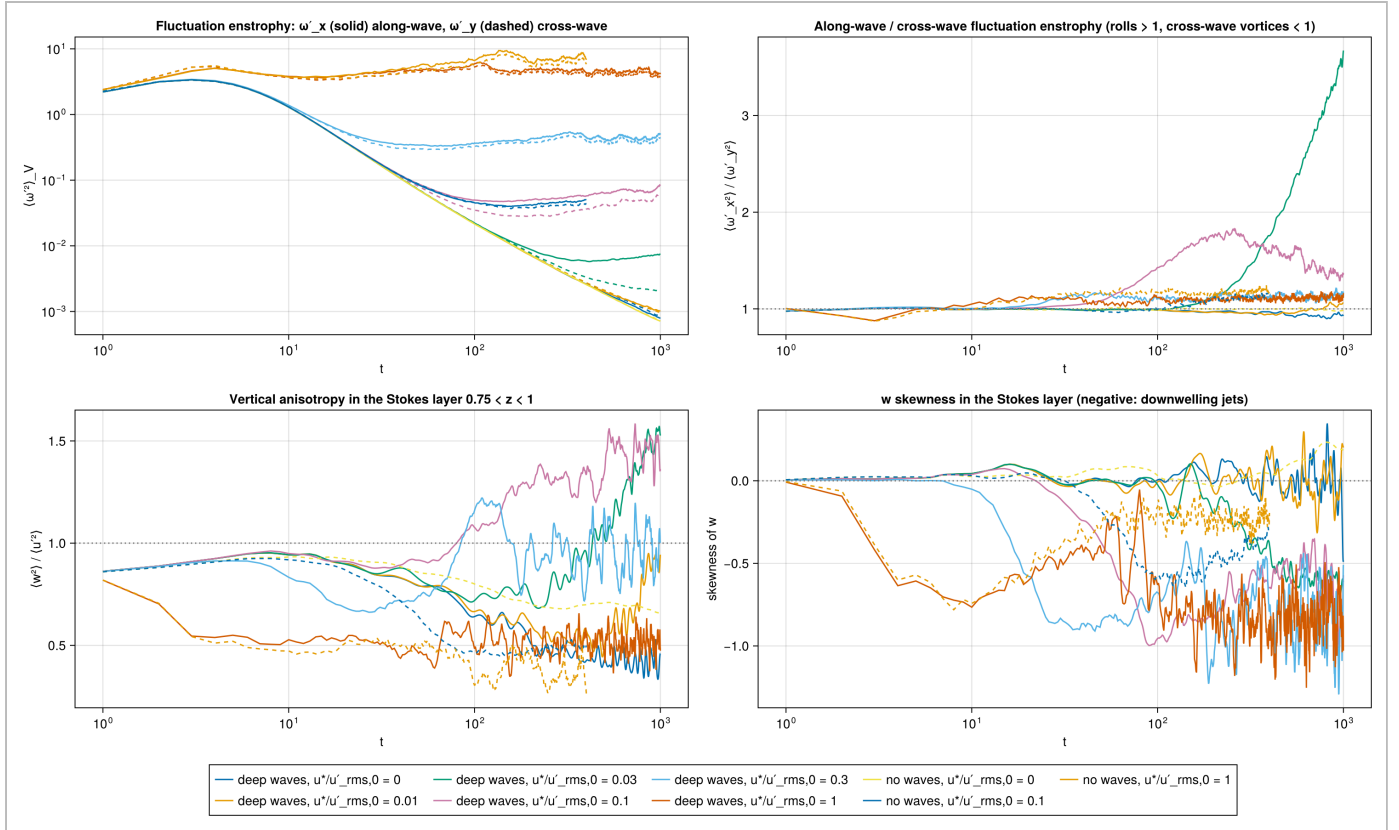
Deep-wave Stokes drift, stress applied along the waves at the surface, decaying turbulence from a shear-free Lagrangian field. Because the fluctuation velocity $u'_{rms}(t)$ decays while τ is fixed, $u^*/u'_{rms}(t)$ increased in every stressed run over the durations sampled, and it is this instantaneous ratio rather than $u^*/u'_{rms,0}$ that organises the cases. Whether a weak stress eventually crosses the roll threshold is not established by these finite-time runs: the stress itself sustains fluctuations (the enstrophy plateaus), so u'_{rms} need not decay indefinitely.



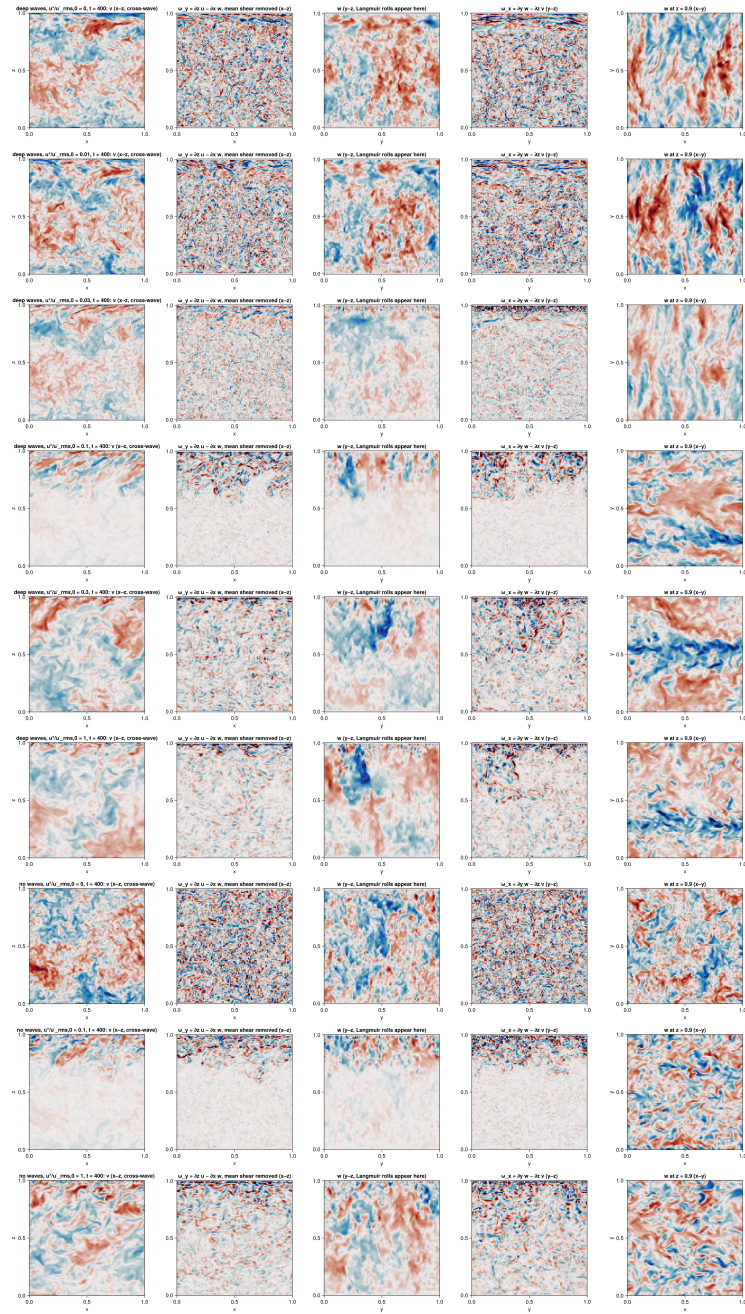
PROVISIONAL – VERY LOW RESOLUTION ($256^2 / 256^3$ and coarser). Regime diagram (128^3 , seed 1). Left: $u^*/u'_{rms}(t)$ ($u'_{rms}(t) = \sqrt{2K_{fluct}}$, the rms of the velocity fluctuation about the horizontal-mean profile) for every stressed case; the ratio grows as the turbulence decays and saturates once the wind layer sets u'_{rms} . Middle: the along/cross fluctuation-entropy ratio $\langle \omega'^2_x \rangle / \langle \omega'^2_y \rangle$ against u^*/u'_{rms} ; in the runs shown, the roll signature (ratio > 1.3) appears only along the $u^*/u'_{rms,0} = 0.03$ (La 0.17) and 0.1 (La 0.31) curves, beyond $u^*/u'_{rms} \approx 0.7-1$, reaching 1.75 and 3.7; $u^*/u'_{rms,0} = 0.3, 1$ and both wave-free controls stay below 1.2 at any ratio. Right: Stokes-layer vertical anisotropy $\langle w^2 \rangle / \langle u'^2 \rangle$ ($0.75 < z < 1$) rises above 1 (to 1.3–1.5) in the same two cases and falls to ≈ 0.5 for the shear-dominated cases. Figure `final_regime.png`.



PROVISIONAL – VERY LOW RESOLUTION ($256^2 / 256^3$ and coarser). Energy budget. Total Lagrangian kinetic energy K/K_0 decays to $1.3e-3$ at $t = 400$ with waves against $0.9e-3$ without (top left); mean-profile energy K_{jet} (top right) grows at zero stress from $t \approx 100$ while the control's falls, and the fraction K_{jet}/K_{rel} (bottom left) reaches 0.40 at $t = 1000$ (control ≤ 0.02). The bulk velocity follows τt exactly (bottom middle, momentum check), and the budget residual $\epsilon = \tau(u)_{top} - dK/dt$ is positive at all times (implicit dissipation only removes energy). Figure `final_energy.png`.



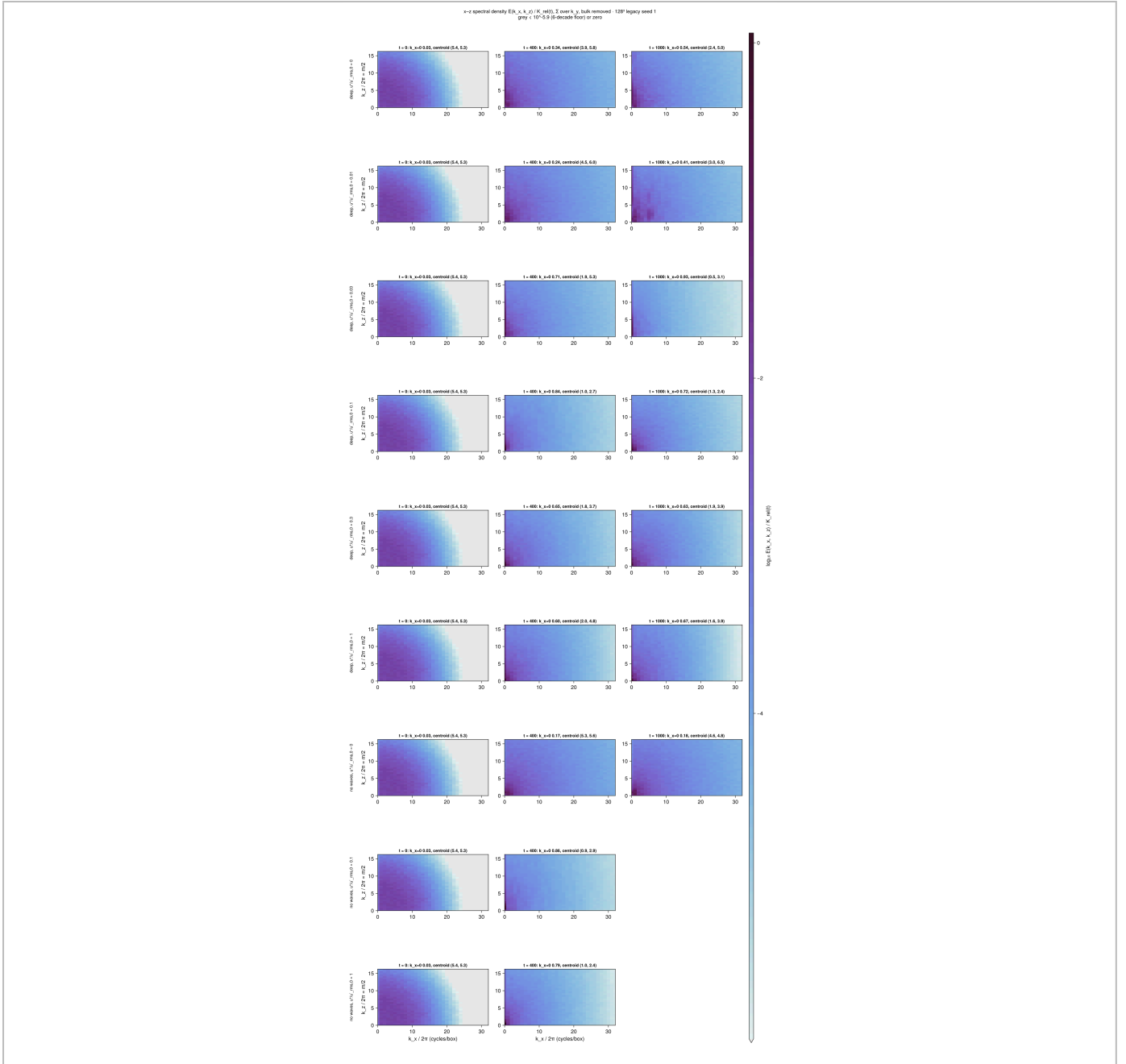
PROVISIONAL – VERY LOW RESOLUTION ($256^2 / 256^3$ and coarser). Structure diagnostics against time. Fluctuation enstrophy (top left) decays as $t^{-1.6}$ without stress and plateaus at a stress-set level otherwise ($r = 0.03$ from $t \approx 300$, 0.1 from ≈ 100 , 0.3 from ≈ 30 , 1 from ≈ 5). The roll ratio (top right) departs from 1 at $t \approx 80$ for $r = 0.1$ and $t \approx 277$ for $r = 0.03$ and keeps rising to 3.7 by $t = 1000$ in the latter; the w skewness turns negative (downwelling plumes) at the same times. Figure final_anisotropy.png.



PROVISIONAL — VERY LOW RESOLUTION ($256^2 / 256^3$ and coarser). Sections at $t = 400$ (rows: deep waves $r = 0, 0.01, 0.03, 0.1, 0.3, 1$; wave-free $r = 0, 0.1, 1$; columns: cross-wave v in x - z , ω_y with the mean shear removed, w in y - z , ω_x , and w at $z = 0.9$). At $r = 0$ and 0.01 , w at $z = 0.9$ is organised into cross-wave (y -elongated) streaks and the y - z vorticity shows no surface-attached cells; at $r = 0.1$ the y - z plane shows surface-attached counter-rotating cells with narrow downwelling plumes reaching $z \approx 0.6$ and w at $z = 0.9$ is x -elongated (Langmuir rolls); at $r = 0.3$ and 1 the wind layer fills the column and the structures match the wave-free controls at the same stress. Figure `final_sections_t400.png`.



PROVISIONAL – VERY LOW RESOLUTION ($256^2 / 256^3$ and coarser). Jet metrics. Interior sign reversals of $U_L - U_{\text{bulk}}$ settle at 3 for $r = 0$ and 0.01 by $t \approx 300$ (no reversals for $r \geq 0.1$ once the wind layer forms), the band wavelength reaches ≈ 0.75 at $t = 1000$, the same order as the local Rhines estimate $2\pi\sqrt{(U/|\partial z z u^s|)}$ (dashed; 0.18 at $z = 0.9$, 0.32 at $z = 0.75$) without being a quantitative match, and the relative profile becomes persistent (10-unit autocorrelation ≈ 1) after $t \approx 50$. Figure `final_jets.png`.



PROVISIONAL — VERY LOW RESOLUTION ($256^2 / 256^3$ and coarser). Directional spectra in the x-z plane. $E(k_x, k_z)/K_{rel}(t)$ of the velocity fluctuation about the volume mean, summed over k_y (Fourier in x, cosine/sine in the bounded z direction), at $t = 0, 400$ and 1000 for the 128^3 seed-1 deep-wave sweep and the wave-free controls (the legacy 128^3 draw, a different realization from the 256^3 runs). One cell per mode on axes in cycles per box ($k_z/2\pi = m/2$), equal aspect; $\log_{10}$ colour shared by all panels, six decades, grey below the floor. Energy collects in $k_x = 0$ modes as the mean profile organises: fraction 0.34 ($t = 400$) and 0.54 ($t = 1000$) at zero stress against 0.17 – 0.18 without waves; $0.71 \rightarrow 0.93$ at $u^*/u'_{rms,0} = 0.03$; 0.84 at 0.1 ($t = 400$). The spectra sum exactly to the relative kinetic energy. The earlier panels in `final_spectra.png` used a DCT/DST scaling that summed to half the energy (fractions and centroids were unaffected). Figure `final_xz_spectra_heatmap.png`.

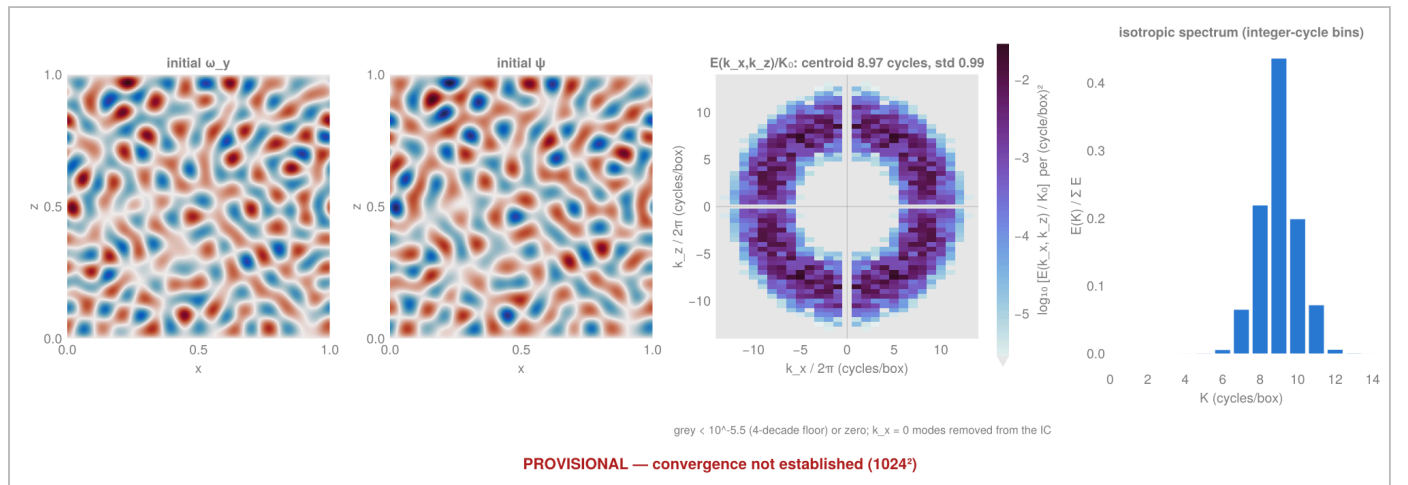
Empirical brackets from a sampled sweep, not a universal transition. The observed pattern — rolls for $u^*/u'_{rms}(t) \gtrsim 0.7$ – 1 with $La \lesssim 0.3$, persisting WC25 jets while $u^*/u'_{rms} \lesssim 0.3$, shear turbulence for $La \gtrsim 0.5$ — rests on one seed at $r = 0.01, 0.1$ and 0.3 , two seeds at $0, 0.03$ and 1 , and two grids at $0, 0.03$ and 0.1 . It is a description of these runs, not a necessary or sufficient condition. The onset time differs between grids ($r = 0.1$: $t = 82$ at 128^3 , 156 at 256^3) while the onset ratio agrees (1.02 vs 1.06); agreement between two grids is not a demonstration of convergence. *Corrected 2026-10-03*: an earlier version said the $r = 0.03$ onset was less certain at 256^3 because a layer-restricted measure gave $t \approx 270$ while the “raw volume ratio $\langle \omega_x'^2 \rangle / \langle \omega_y'^2 \rangle$ ” was only ≈ 1.2 at $t = 400$ against 1.75 at 128^3 . That compared different quantities: the 1.2 (256^3) is the raw ratio of total vorticity, R_{raw} , which includes the mean shear (128^3 : 1.00), whereas the 1.75 (128^3) is the fluctuation ratio R_{fl} (256^3 : 1.66), and $t \approx 270$ is the R_{fl} onset (definitions under Editorial corrections). With the same measure on both

grids the $r = 0.03$ onset agrees (R_{ff} : $t = 277$ and 270). The 128^3 and 256^3 runs compared here started from different realizations; in the interim same-field comparison (REPORT_resolution.md) the $r = 0.1$ onset is $t = 84$ (128^3) and 156 (256^3), so that shift is a resolution effect; 512^3 is pending. The 128^3 sweep is strongly dissipative ($K/K_0 = 2.7\text{e-}3$ by $t = 190$), the jets are 3–10 $\times$ weaker than WC25's spun-up cases, and the box holds only 2–3 bands. At $r \geq 0.3$ the mean advection reaches $O(1)$ and the implicit WENO dissipation becomes velocity-dependent (a Galilean-invariance caveat for the LES, not a CFL-stability problem: the adaptive step kept CFL at 0.5). The 256^3 extension to $t = 1000$ at $r = 0.03$ is now complete (see the interim report). The 512^3 checks, additional stress values and seeds, and domain-sensitivity tests remain relevant before stating quantitative thresholds.

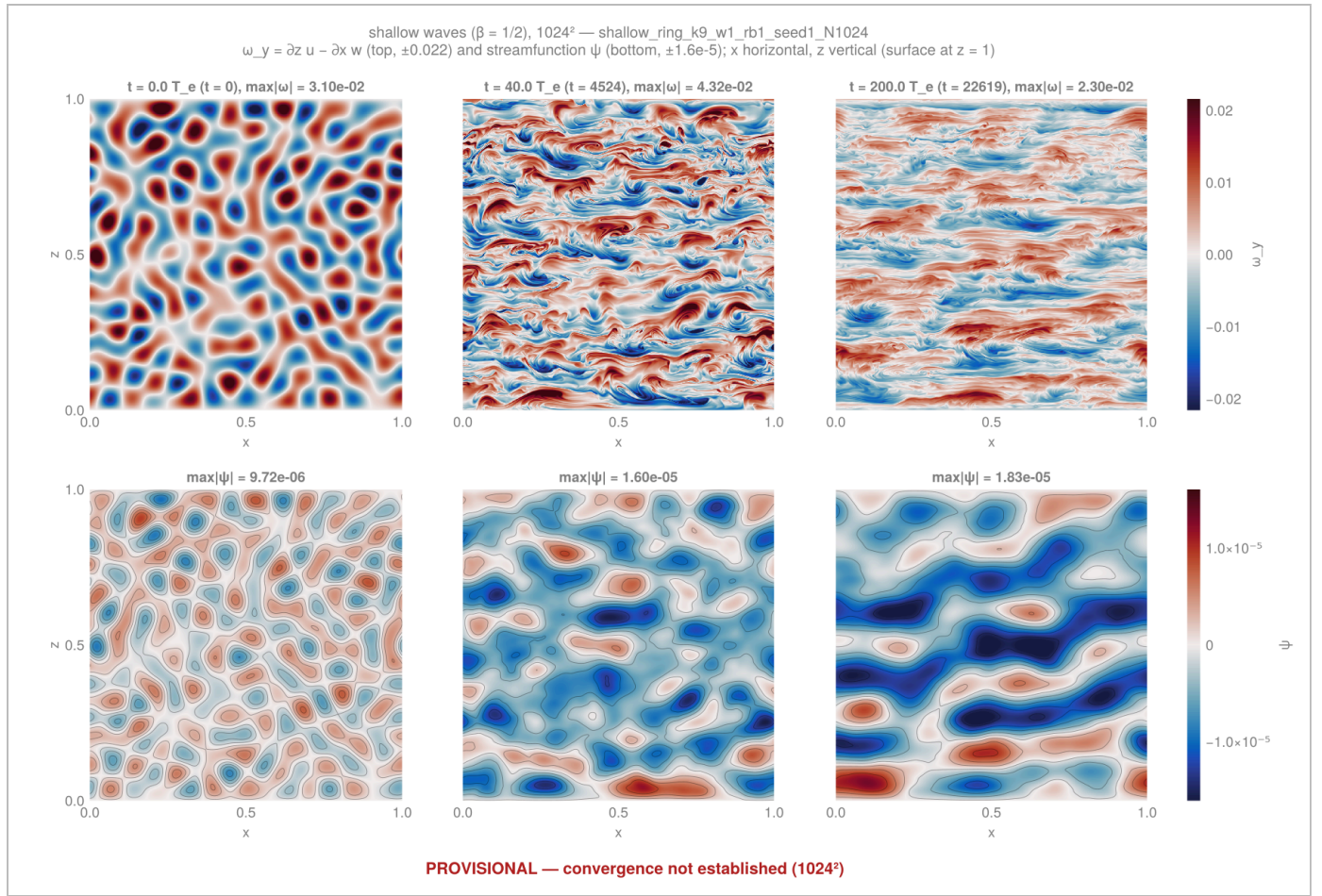
B • Vallis narrow-ring decay in two-dimensional WC25 flow

One Lagrangian ring field ($k_0 = 9$ cycles/box, $\sigma =$ one mode spacing, $rb = 1$, seed 1) evolved under shallow waves, deep waves and no waves; the ring parameters follow the Vallis demo code (its amplitude law $1/[K^2 + (K^2 - 3200)^2]$ peaks at 9.0 cycles and, once its unnormalised FFT is undone, corresponds to $rb \approx 0.96$) and VM93 ($\beta = 400$, $E = 1$, $K = 12$ give $rb \approx 1.96$ under the assumed source normalization; their $Z/E = 2K^2$ is consistent with a narrow ring but does not determine its width, so $\sigma = 1$ mode is a chosen value tested against $\sigma = 0.5$ and 2). The comparison with the published figures is qualitative and source-matched, not an exact reproduction.

Resolution of the main figures (updated 2026-10-04). Every main figure in this section uses 1024^2 , the highest grid completed for these cases (ring-9 families to 200 T_e; $k_0 = 12$ ensemble to 60 T_e). The earlier 256^2 versions are retained only as labelled lower-resolution baselines inside explicit comparisons (three-grid figures res3v2 and ens12). Two figures remain at 256^2 because their cases exist only there: the doubly periodic β -plane comparison and the sensitivity variants; both are labelled as comparisons.

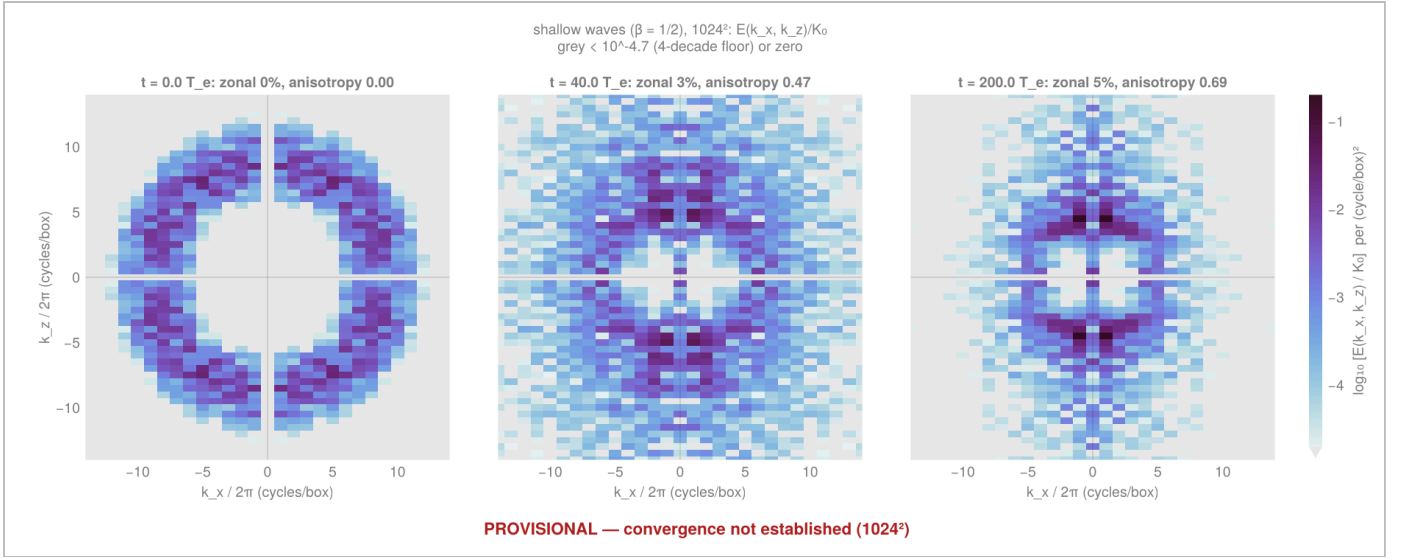


PROVISIONAL — convergence not established (1024²). Shared initial condition (1024²): ω_y , ψ , $E(k_x, k_z)/K_0$ and the isotropic spectrum. Centroid 8.97 cycles, standard deviation 0.99, $w = \omega_y = 0$ on the walls, anisotropy index 0.004, $k_x = 0$ modes removed. $u'_{\text{rms},0} = 1.56\text{e-}4$, $K = 1.22\text{e-}8$, enstrophy $7.87\text{e-}5$. Heatmap: spectral density per (cycle/box)², normalized by the initial energy K_0 (Σ density $\times$ cell area = $K(t)/K_0$; checked to $1\text{e-}8$ at every panel time), one cell per lattice mode ($k_x/2\pi$ integer, $k_z/2\pi = n/2$ for the wall sine basis), $\log_{10}$ colour on a perceptually uniform sequential map with limits shared across the panels and families being compared; cells more than four decades below the shared maximum, or exactly zero, are grey. No interpolation, no overlays. *Provenance*: 1024² run ring_k9_w1_rb1_seed1_N1024 (COMPLETED to 200 T_e), same physical parameters and the same continuous initial coefficients as the earlier 256^2 run ($t = 0$ spectral density agrees to 6×10^{-4} ; main1024_provenance.txt). Lower-resolution versions appear only in the explicit three-grid comparison res3v2 ($256^2 / 512^2 / 1024^2$). Figure main1024_ic_heatmap.png.

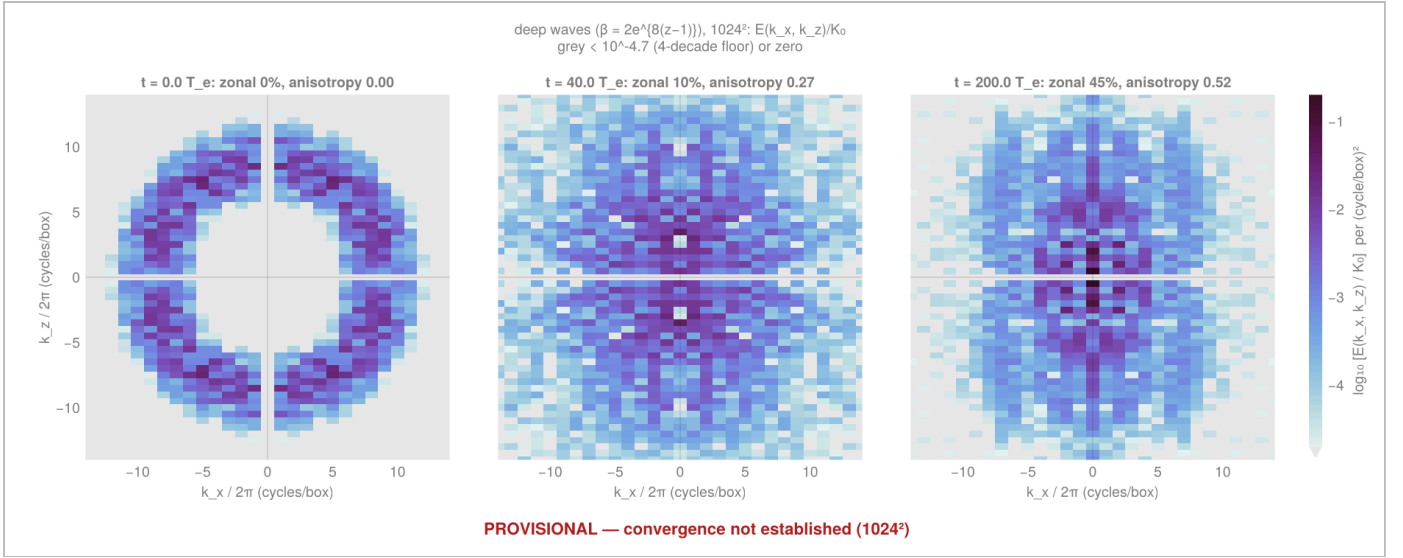


PROVISIONAL — convergence not established (1024²). Shallow waves ($\beta = 1/2$), the Fig. 12.4 analogue (1024²). ω_y (top, fixed limits ± 0.022) and ψ (bottom, $\pm 1.6e-5$) at 0, 40 and 200 T_e . The isotropic blob pattern becomes x-elongated streaks by 40 T_e and bands by 200 T_e ; ψ organises into x-elongated cells with 4–5 alternations across the depth. Energy changes by -0.003% and enstrophy decays to 43% over 200 T_e . *Provenance:* 1024² run ring_k9_w1_rb1_seed1_N1024 (COMPLETED to 200 T_e), same physical parameters and the same continuous initial coefficients as the earlier 256² run ($t = 0$ spectral density agrees to 6×10^{-4} ; main1024_provenance.txt). Lower-resolution versions appear only in the explicit three-grid comparison res3v2 (256² / 512² / 1024²). Figure main1024_panels_shallow.png.

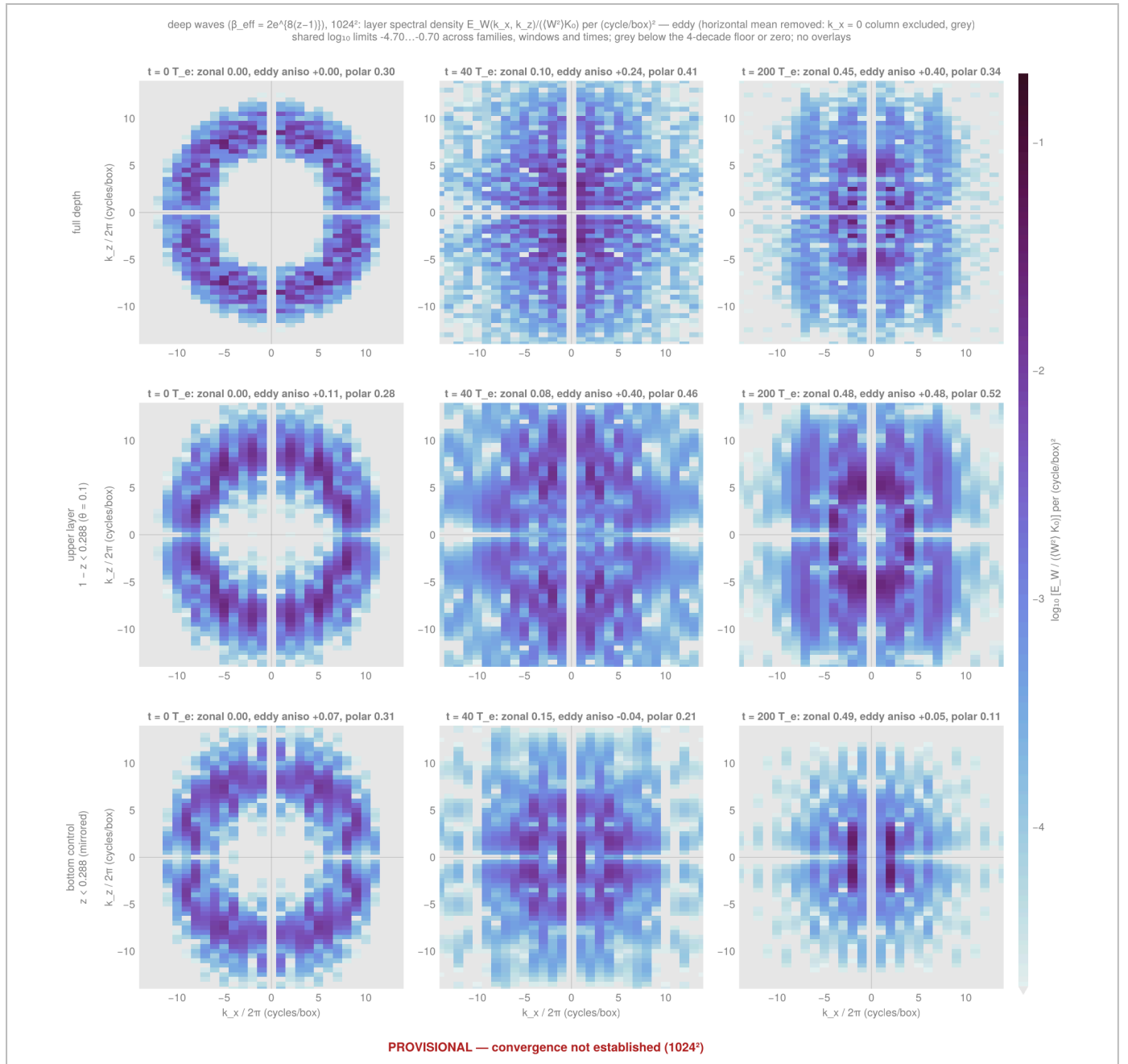
Overlay audit (2026-10-04). No main spectrum in this section draws a Rhines, Vallis Eq. 12.14 or any other theoretical curve; the replacement figures are regenerated at 1024² without overlays, and no curve was shrunk or fitted to the observed hole. A global velocity-scale estimate $K^2 = \beta |\cos \theta| / u'_{\text{rms}}(t)$ is a reference scale, not the observed boundary of the depleted region, and the ε -based Eq. 12.14 curve is unsupported because the sampled fluxes give no defensible inverse transfer rate. The numeric audit (measured at 256²) is in REPORT_overlay_audit.md.



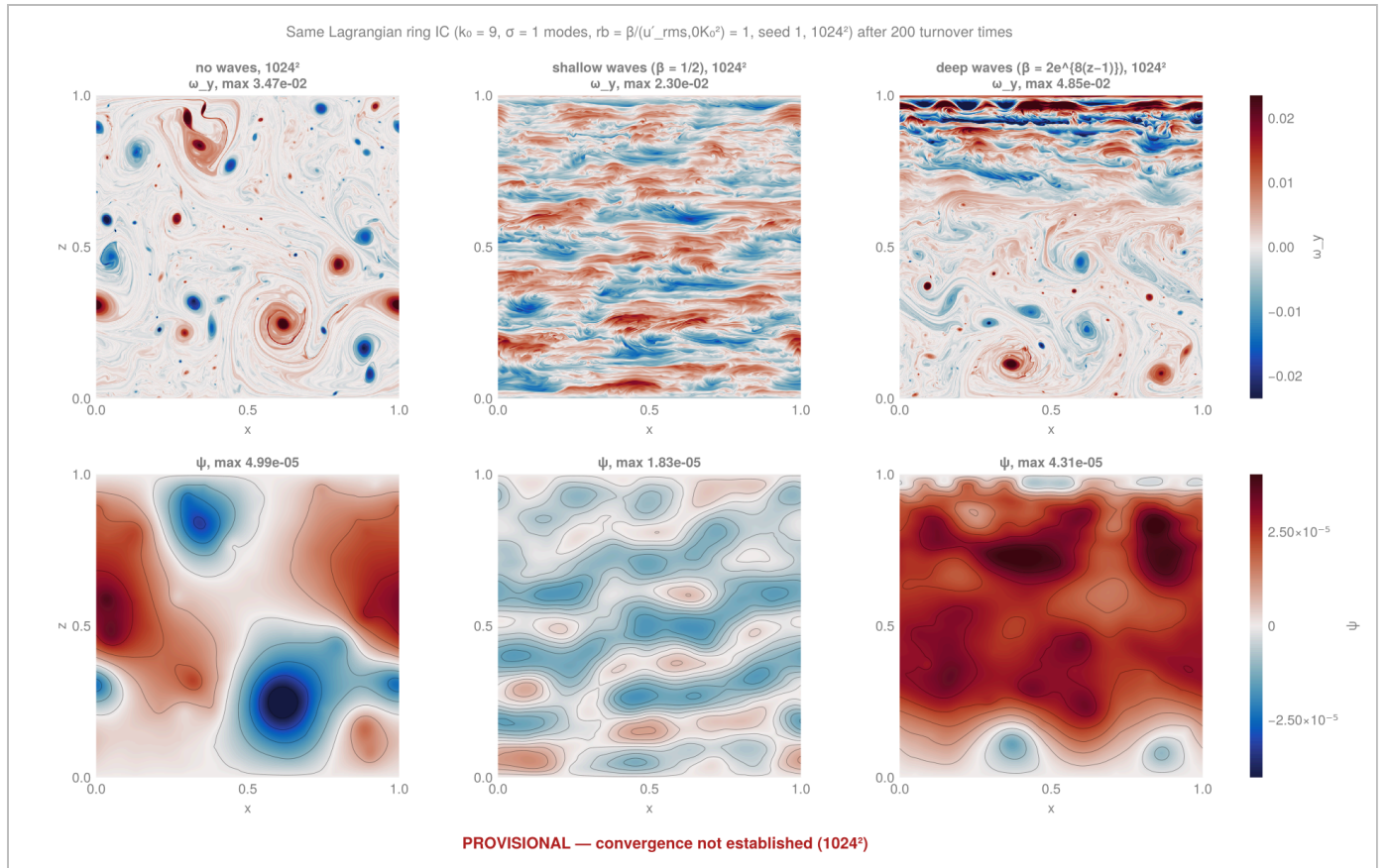
PROVISIONAL — convergence not established (1024²). Shallow waves, the Fig. 12.3 analogue (1024²). $E(k_x, k_z)/K_0$ at 0, 40 and 200 T_e . The energy leaves the ring and concentrates off the $k_x = 0$ axis near the k_z axis ($|k_x| \leq 3$, $k_z \approx \pm 4-6$ cycles at 40 T_e ; $|k_x| \leq 2$, $k_z \approx \pm 4-5$ at 200 T_e), leaving a depleted region around the k_x axis. Anisotropy index 0.00 $\rightarrow$ 0.47 $\rightarrow$ 0.69; exact-zonal fraction 0 $\rightarrow$ 3% $\rightarrow$ 5%. No Rhines or other estimated boundary is drawn (see the overlay audit above). Shared $\log_{10}$ limits -4.70 to -0.70 for the shallow, deep and wave-free spectra. Heatmap: spectral density per (cycle/box)², normalized by the initial energy K_0 (Σ density $\times$ cell area = $K(t)/K_0$; checked to $1e-8$ at every panel time), one cell per lattice mode ($k_x/2\pi$ integer, $k_z/2\pi = n/2$ for the wall sine basis), $\log_{10}$ colour on a perceptually uniform sequential map with limits shared across the panels and families being compared; cells more than four decades below the shared maximum, or exactly zero, are grey. No interpolation, no overlays. *Provenance*: 1024² run ring_k9_w1_rb1_seed1_N1024 (COMPLETED to 200 T_e), same physical parameters and the same continuous initial coefficients as the earlier 256² run ($t = 0$ spectral density agrees to 6×10^{-4} ; main1024_provenance.txt). Lower-resolution versions appear only in the explicit three-grid comparison res3v2 (256² / 512² / 1024²). Figure main1024_spectra_shallow_heatmap.png.



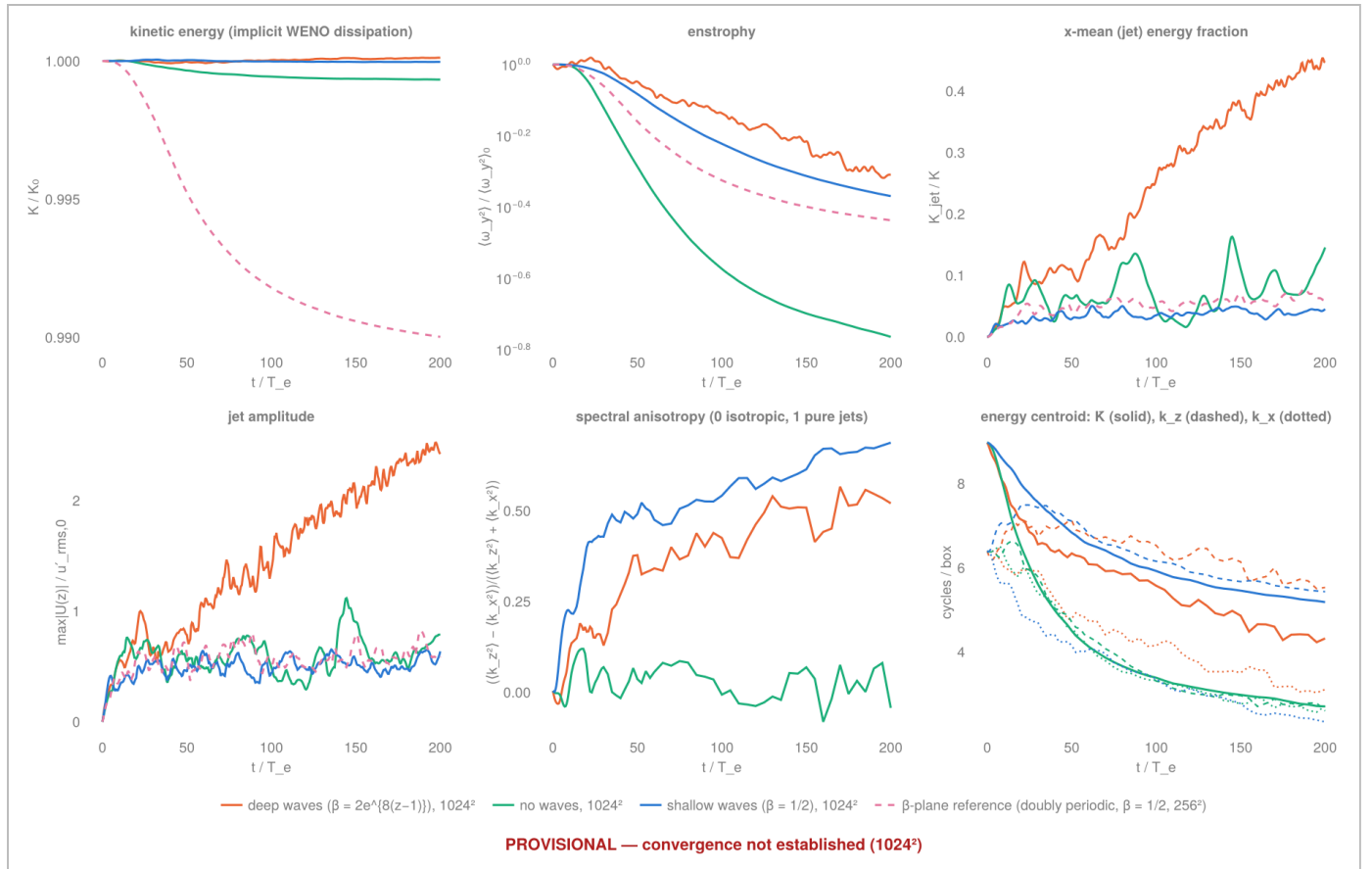
PROVISIONAL — convergence not established (1024²). Deep waves, same diagnostic (1024², same ring initial condition, seed 1) at 0, 40 and 200 T_e, on the same colour scale as the shallow-wave figure ($\log_{10}$ limits -4.70 to -0.70 , shared with the wave-free spectra), so the figures compare cell by cell. The spectrum is a Fourier (x) $\times$ sine (z) decomposition of the streamfunction over the whole column between the walls: $k_z/2\pi = n/2$ counts vertical half-wavelengths, the $k_z = 0$ row is empty (no $n = 0$ sine mode), and each cell mixes contributions from every depth. $\beta_{\text{eff}}(z) = \partial_{zz} u^* = 2e^{8(z-1)}$ varies by a factor of about 3000 over the depth, so no constant- β estimate applies. Energy leaves the ring and concentrates in and next to the exact-zonal $k_x = 0$ column at low $|k_z|$: exact-zonal fraction $0 \rightarrow 10\% \rightarrow 45\%$ and anisotropy $0.00 \rightarrow 0.27 \rightarrow 0.52$, whereas shallow waves leave only 5% exact-zonal (anisotropy 0.69) at 200 T_e. Single realization; the deep-wave zonal fraction differs between grids (0.39 at 256², 0.38 at 512², 0.45 at 1024²), so convergence is not established. Wave-free spectra: main1024_spectra_none_heatmap.png. Heatmap: spectral density per (cycle/box)², normalized by the initial energy K_0 (Σ density $\times$ cell area = $K(t)/K_0$; checked to $1e-8$ at every panel time), one cell per lattice mode ($k_x/2\pi$ integer, $k_z/2\pi = n/2$ for the wall sine basis), $\log_{10}$ colour on a perceptually uniform sequential map with limits shared across the panels and families being compared; cells more than four decades below the shared maximum, or exactly zero, are grey. No interpolation, no overlays. *Provenance*: 1024² run ring_k9_w1_rb1_seed1_N1024 (COMPLETED to 200 T_e), same physical parameters and the same continuous initial coefficients as the earlier 256² run ($t = 0$ spectral density agrees to 6×10^{-4} ; main1024_provenance.txt). Lower-resolution versions appear only in the explicit three-grid comparison res3v2 (256² / 512² / 1024²). Figure main1024_spectra_deep_heatmap.png.



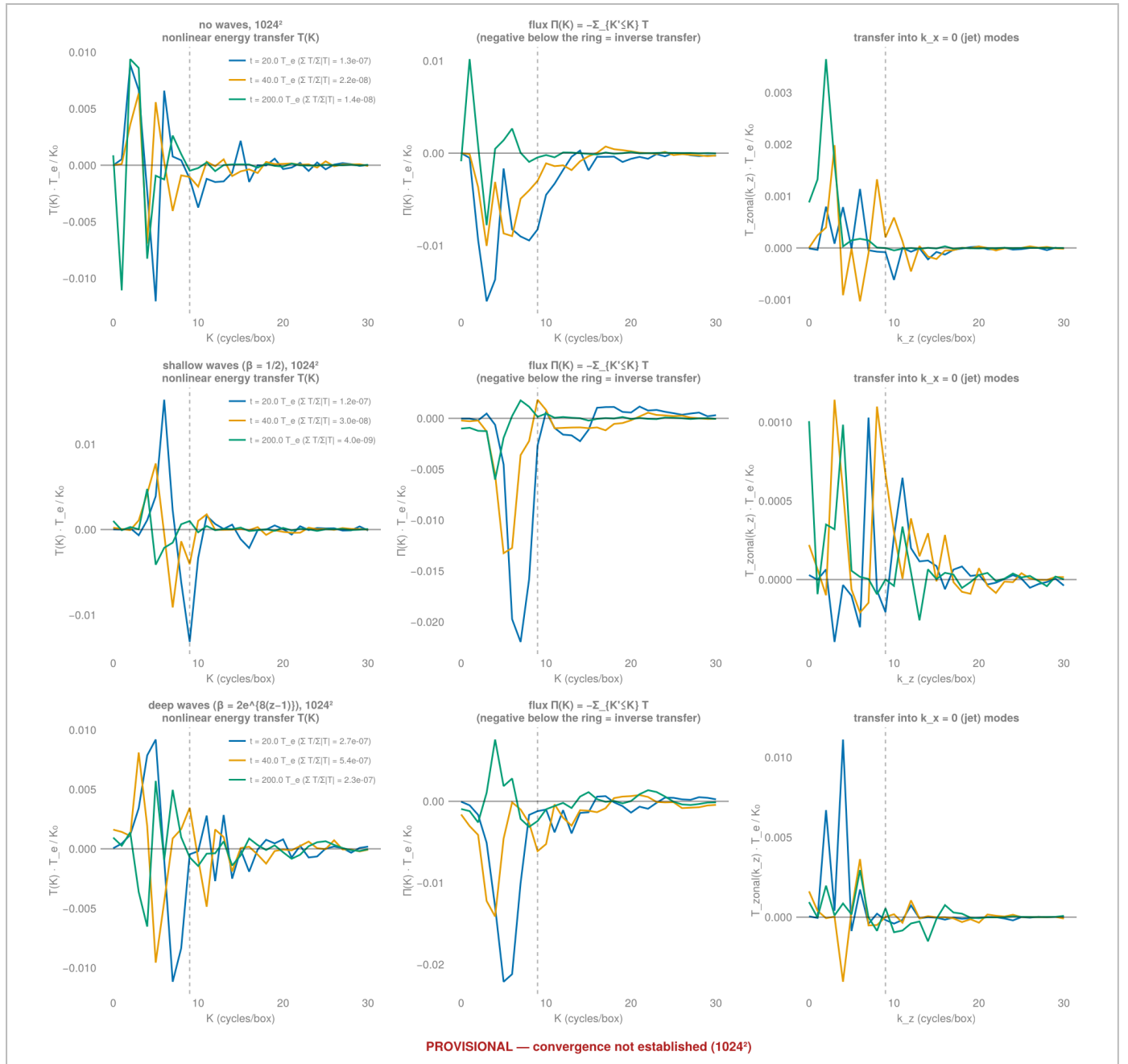
PROVISIONAL — convergence not established (1024²). Is a dumbbell hidden in the deep-wave full-depth average? (new analysis, 2026-10-04) Eddy kinetic-energy spectra (horizontal mean removed) of the 1024² deep-wave run over the full depth (top), the upper layer where $\beta_{\text{eff}} \geq 0.1 \beta_{\text{eff}}(1)$, i.e. $1 - z < 0.288$ ($\theta = 0.1$) (middle), and a mirrored bottom layer of the same thickness as a control (bottom), at 0, 40 and 200 T_e . The layer was chosen beforehand from the prescribed $\beta_{\text{eff}}(z) = 2e^{8(z-1)}$; it is an operational choice, not a sharp boundary. Actual velocities are tapered smoothly to zero at the artificial cut and left untouched at the real wall, and transformed on the full domain, so the wavenumber units match the main spectra; Parseval holds to $1e-10$. A thin layer cannot resolve k_z finely: the window smears k_z by about ± 1 cycle/box (side lobes near 3 cycles/box cause the vertical streaks), and features narrower than about 2 cycles/box are not resolved. *Outcome: partial and ambiguous.* In the upper layer the eddy spectrum is more anisotropic than the full-depth average (+0.48 vs +0.40 at 200 T_e ; window baseline on the isotropic initial field +0.11) and concentrates in lobes near the k_z axis ($|k_x| \leq 3$, $k_z \approx \pm 4-6$), robustly across cutoffs $\theta = 0.2, 0.1, 0.05$ and three tapers, and not in the bottom control (+0.05) or the wave-free run (-0.07). A depleted region around the k_x axis, the defining feature of a dumbbell, is not clearly present: 25% of the band energy lies within 30° of the k_x axis, against 8% for shallow waves under the identical window. β_{eff} also varies tenfold inside the layer, so no single dumbbell scale applies. No Rhines or other theoretical curve is drawn. Same colour scale and normalization as the comparison figures: deep / shallow / wave-free at 200 T_e , windows, bandwidth and diagnostics, cutoff and taper sensitivity, total (including the x-mean). Methodology, checks and limitations: REPORT_layer_spectra.md. Figure layer1024_deep_eddy.png.



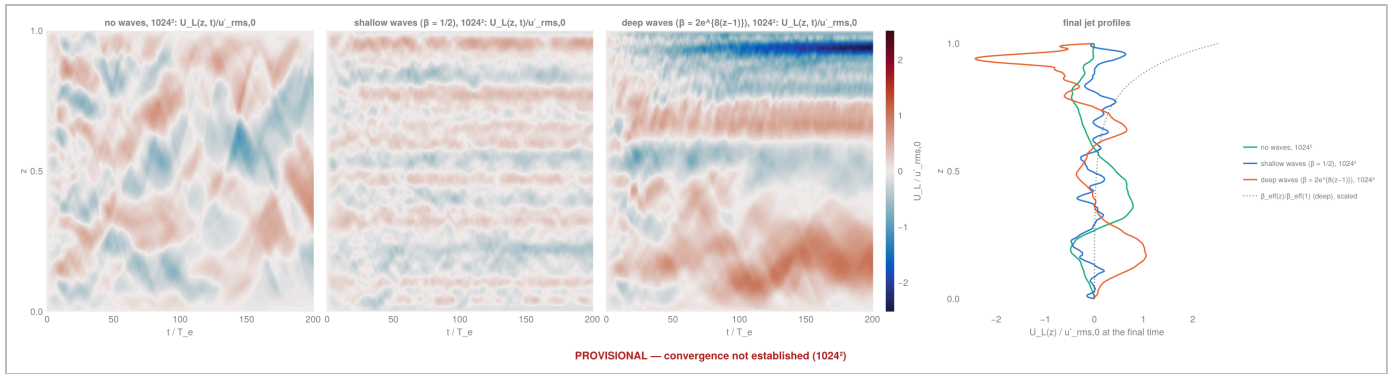
PROVISIONAL — convergence not established (1024²). Three families at 200 T_e (1024²; ω_y top, ψ bottom, shared limits). No waves: Vallis Fig. 11.8 decay — coherent vortices, merger, energy centroid 2.7 cycles, enstrophy 17%. Shallow: x-elongated bands, centroid 5.2 cycles, enstrophy 43%. Deep: β dynamics in the upper part of the column, 2-D vortices below $z \approx 0.6$, a strong surface band structure and a mean streamfunction of one sign carrying the x-mean jet (centroid 4.3 cycles, enstrophy 49%). *Provenance:* 1024² run ring_k9_w1_rb1_seed1_N1024 (COMPLETED to 200 T_e), same physical parameters and the same continuous initial coefficients as the earlier 256² run ($t = 0$ spectral density agrees to 6×10^{-4} ; main1024_provenance.txt). Lower-resolution versions appear only in the explicit three-grid comparison res3v2 (256² / 512² / 1024²). Figure main1024_comparison_t200.png.



PROVISIONAL — convergence not established (1024²). Time series (1024²; the dashed β -plane reference is the 256² doubly periodic run, the only grid at which it exists, and is labelled as such). K/K_0 changes by -0.003% (shallow), -0.07% (none) and $+0.012\%$ (deep) over $200 T_e$, while the β -plane reference loses 1% to hyperviscosity. Enstrophy falls to 43% (shallow), 49% (deep) and 17% (none). K_{jet}/K reaches 0.045 (shallow), 0.145 (none, box-scale condensate, intermittent) and 0.45 (deep); $\max|U_L|/u'_{\text{rms},0}$ reaches 0.64, 0.79 and 2.4. The anisotropy index rises to 0.69 (shallow) and 0.52 (deep) and stays ≈ 0 without waves. The energy centroid falls from 9 to 5.2 (shallow), 4.3 (deep) and 2.7 cycles (none). *Provenance:* 1024² run ring_k9_w1_rb1_seed1_N1024 (COMPLETED to 200 T_e), same physical parameters and the same continuous initial coefficients as the earlier 256² run ($t = 0$ spectral density agrees to 6×10^{-4} ; main1024_provenance.txt). Lower-resolution versions appear only in the explicit three-grid comparison res3v2 (256² / 512² / 1024²). Figure main1024_timeseries.png.



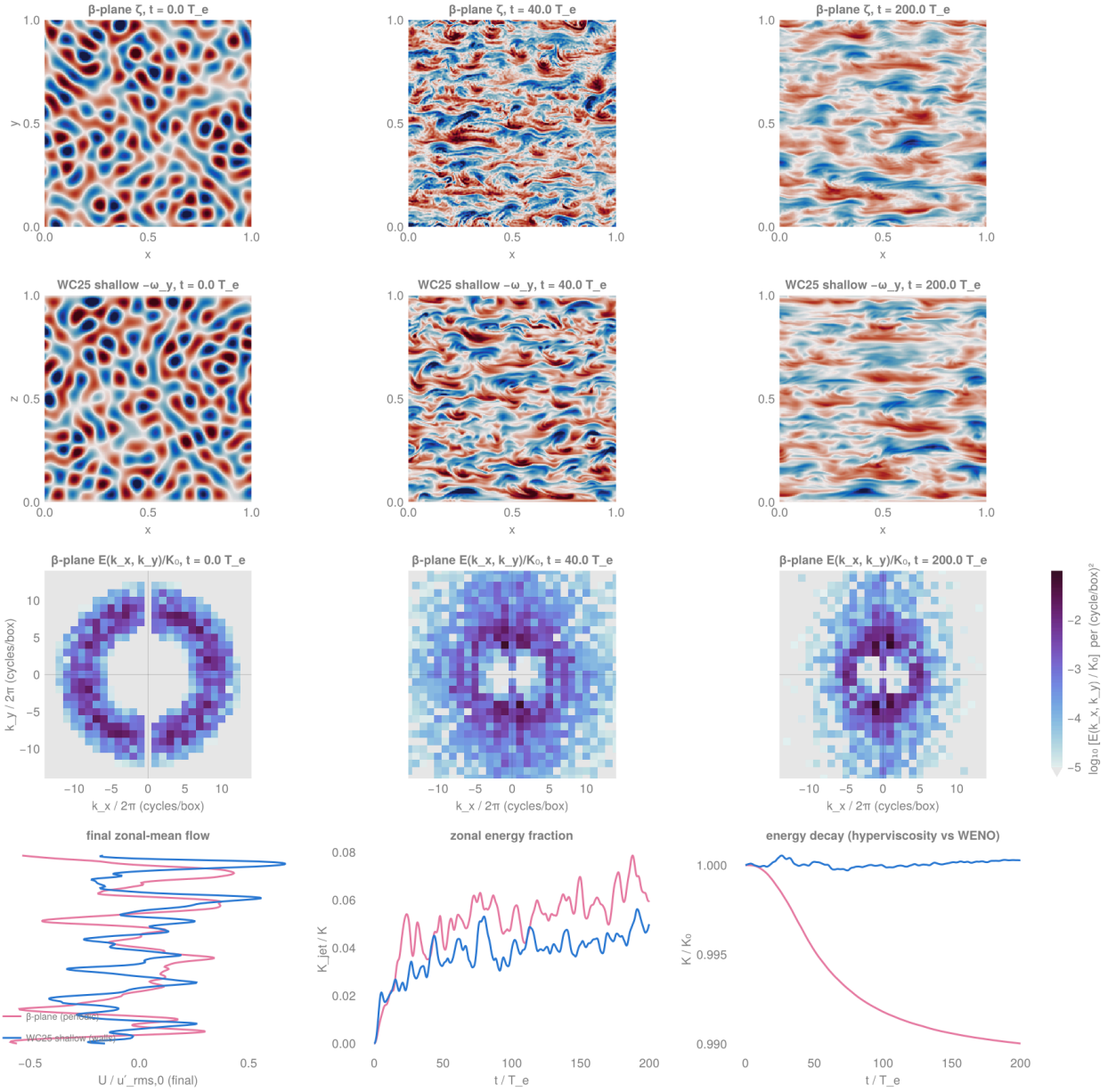
PROVISIONAL — convergence not established (1024²). Spectral transfer (1024²). With waves, $T(K)$ is negative at the ring ($K \approx 7-9$) and positive at smaller K , so the flux $\Pi(K) = -\sum_{K' \leq K} T$ is negative below the ring: an inverse energy transfer, strongest at $20 T_e$ and weakening by $200 T_e$ ($\sum T/\sum |T| \leq 6e-7$, so the diagnostic is conservative). The transfer into $k_x = 0$ modes is about $10\times$ larger with deep waves than with shallow waves and concentrated at $k_z \approx 2-4$ cycles. *Provenance:* 1024² run ring_k9_w1_rb1_seed1_N1024 (COMPLETED to $200 T_e$), same physical parameters and the same continuous initial coefficients as the earlier 256² run ($t = 0$ spectral density agrees to 6×10^{-4} ; main1024_provenance.txt). Lower-resolution versions appear only in the explicit three-grid comparison res3v2 (256² / 512² / 1024²). Figure main1024_transfer.png.



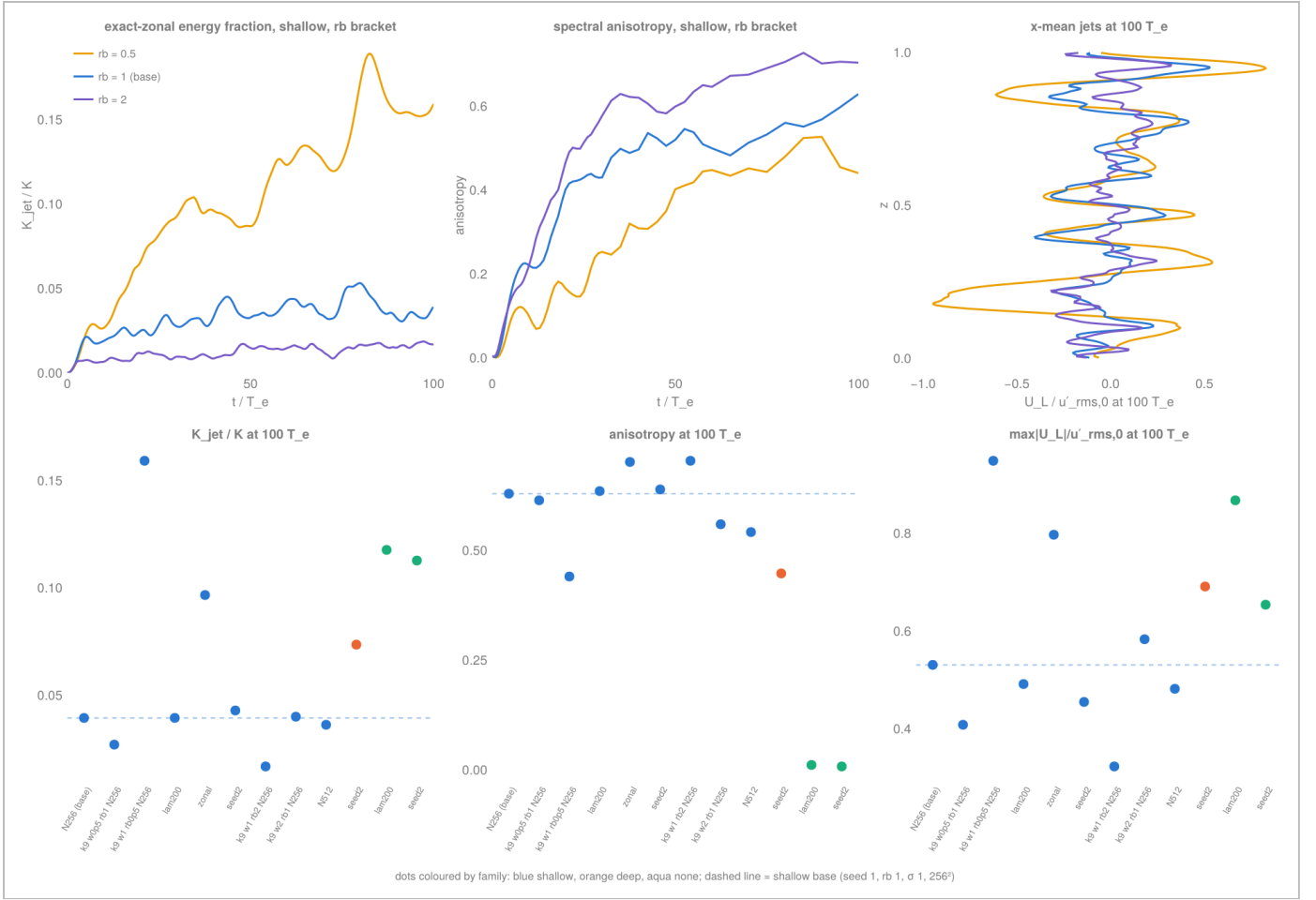
PROVISIONAL — convergence not established (1024²). x-mean jets (1024²). $U_L(z, t)/u'_{rms,0}$ for no waves (left), shallow waves (middle) and deep waves (third), with final profiles at right. Shallow: many fine, weak bands (max 0.64 $u'_{rms,0}$, 18 sign reversals, wavelength ≈ 0.075). Deep: a surface jet toward $-x$ at $z \approx 0.9$ reaching $-2.4 u'_{rms,0}$ and interior jets toward $+x$ near $z \approx 0.65$ and 0.2 (5 sign reversals), with K_{jet}/K growing almost linearly to 0.45. *Provenance:* 1024² run ring_k9_w1_rb1_seed1_N1024 (COMPLETED to 200 T_e), same physical parameters and the same continuous initial coefficients as the earlier 256² run ($t = 0$ spectral density agrees to 6×10^{-4} ; main1024_provenance.txt). Lower-resolution versions appear only in the explicit three-grid comparison res3v2 (256² / 512² / 1024²). Figure main1024_jets.png.

Doubly periodic β -plane reference ($\beta = 1/2$) vs WC25 shallow waves with walls

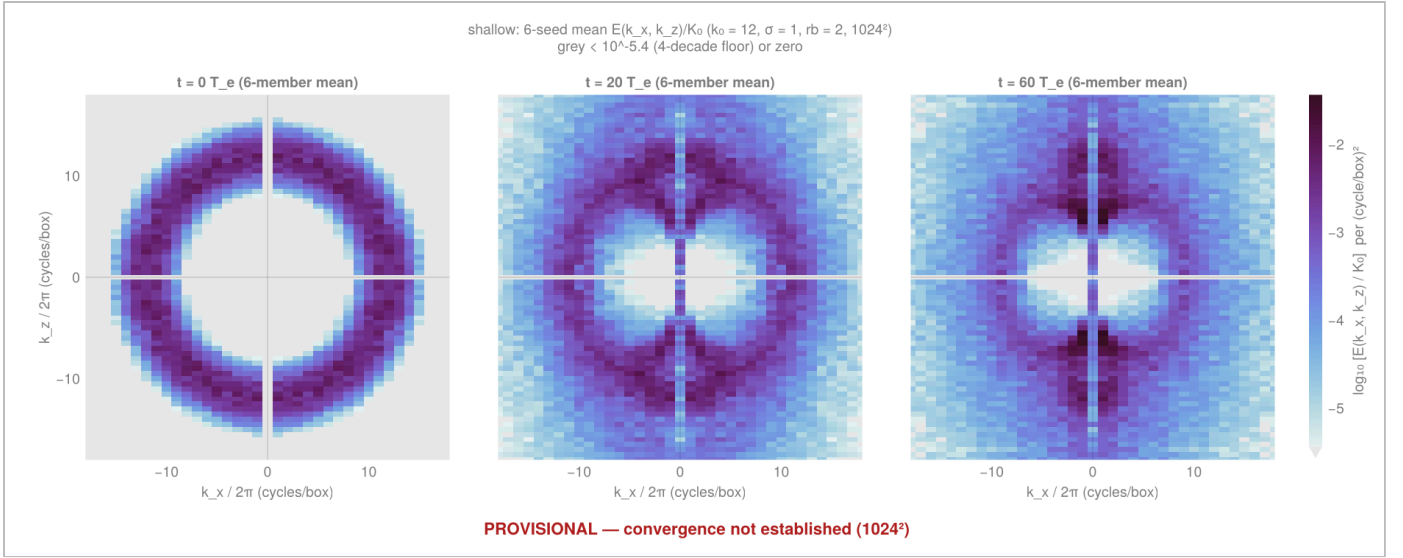
row 3: periodic reference $E(k_x, k_y)/K_0$, integer cells - grey < $10^{-5.0}$ (4-decade floor) or zero



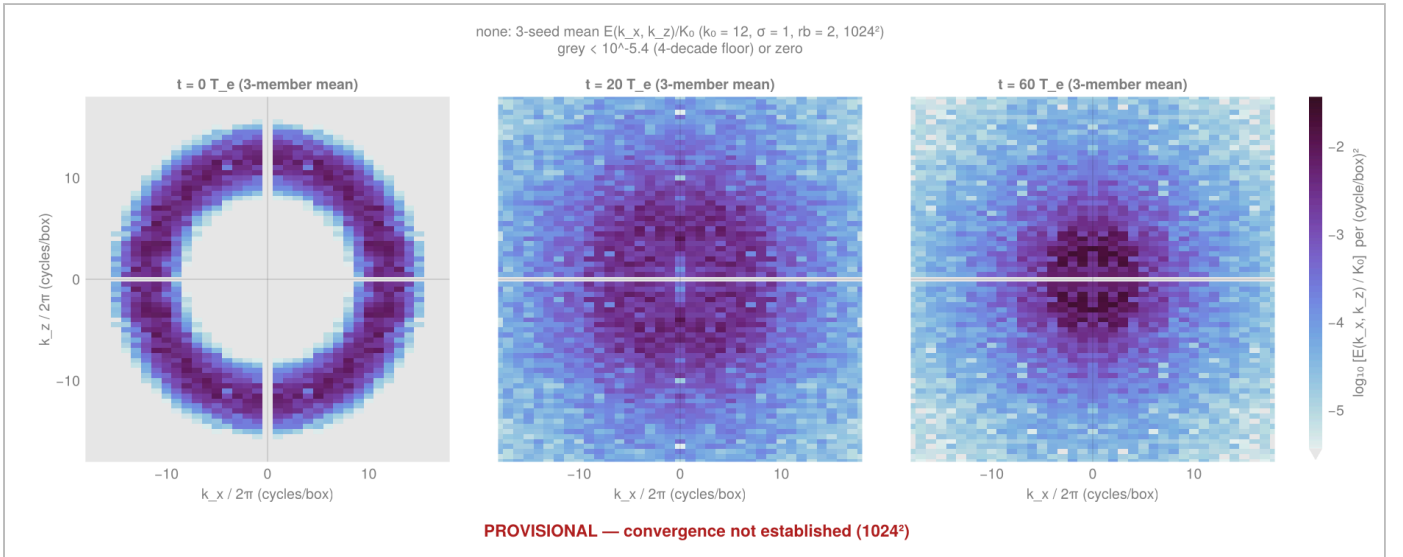
PROVISIONAL — VERY LOW RESOLUTION (256^2); explicit comparison, not a main result. *Resolution:* kept at 256^2 as an explicit comparison because the doubly periodic reference exists only at 256^2 ; both sides are 256^2 so the comparison is matched. The 1024^2 main WC25 shallow case is shown above. **Textbook geometry against walls.** The doubly periodic pseudo-spectral $\beta = 1/2$ reference (same ring law, same $u'_{rms,0}$, RK4, 2/3 dealiasing, ∇^8 hyperviscosity) and the WC25 shallow case with walls give the same vorticity streaks, the same dumbbell spectra and the same exact-zonal fraction (0.060 vs 0.050); the walls are not what limits the jets at $rb = 1$. Energy: -1% (hyperviscosity) vs $+0.03\%$ (WENO). Figure `final_betaplane_heatmap.png`. Heatmap: spectral density per $(\text{cycle/box})^2$, normalized by the initial energy K_0 , one cell per lattice mode. Lattices differ between the two geometries: the doubly periodic reference (unit box, Fourier in both directions) has integer spacing in both $k_x/2\pi$ and $k_y/2\pi$, so its cells are 1×1 $(\text{cycle/box})^2$; the WC25 wall case uses a sine basis in z with $k_z/2\pi = n/2$, i.e. half-integer spacing and $1/2$ $(\text{cycle/box})^2$ cells, so its spectra (other figures) show twice as many rows in k_z , $\log_{10}$ colour on a perceptually uniform sequential map with limits shared across the panels and families being compared; cells more than four decades below the shared maximum, or exactly zero, are grey. No interpolation. The earlier filled-contour version is kept as `final_betaplane.png`.



PROVISIONAL — VERY LOW RESOLUTION (256²); explicit comparison, not a main result. *Resolution:* kept at 256² as an explicit sensitivity comparison because the variants were run only at 256² (plus the 512² base point shown); the 1024² main result for the base case is shown above. **Sensitivity at 100 T_e.** Top: the rb bracket for shallow waves — exact-zonal fraction 0.159 (rb = 0.5), ≈ 0.04 (rb = 1), 0.017 (rb = 2) while the anisotropy index runs the other way (0.44, 0.62, 0.70): a stronger β arrests the cascade nearer the ring and keeps the energy in zonally elongated waves rather than jets. Bottom: every variant at 100 T_e, coloured by family — ring width 0.5–2 modes, 512², seed 2 and the λ = 200 rescaling (u'_{rms,0} = 0.031, identical physics in the nonlinear WENO regime: 0.039 vs ≈ 0.04, with 1% energy loss instead of +0.03%) all sit within the scatter; keeping the k_x = 0 modes adds their 6% almost additively (0.097); deep seed 2 gives 0.074 against ≈ 0.2 for seed 1. Figure final_variants.png.

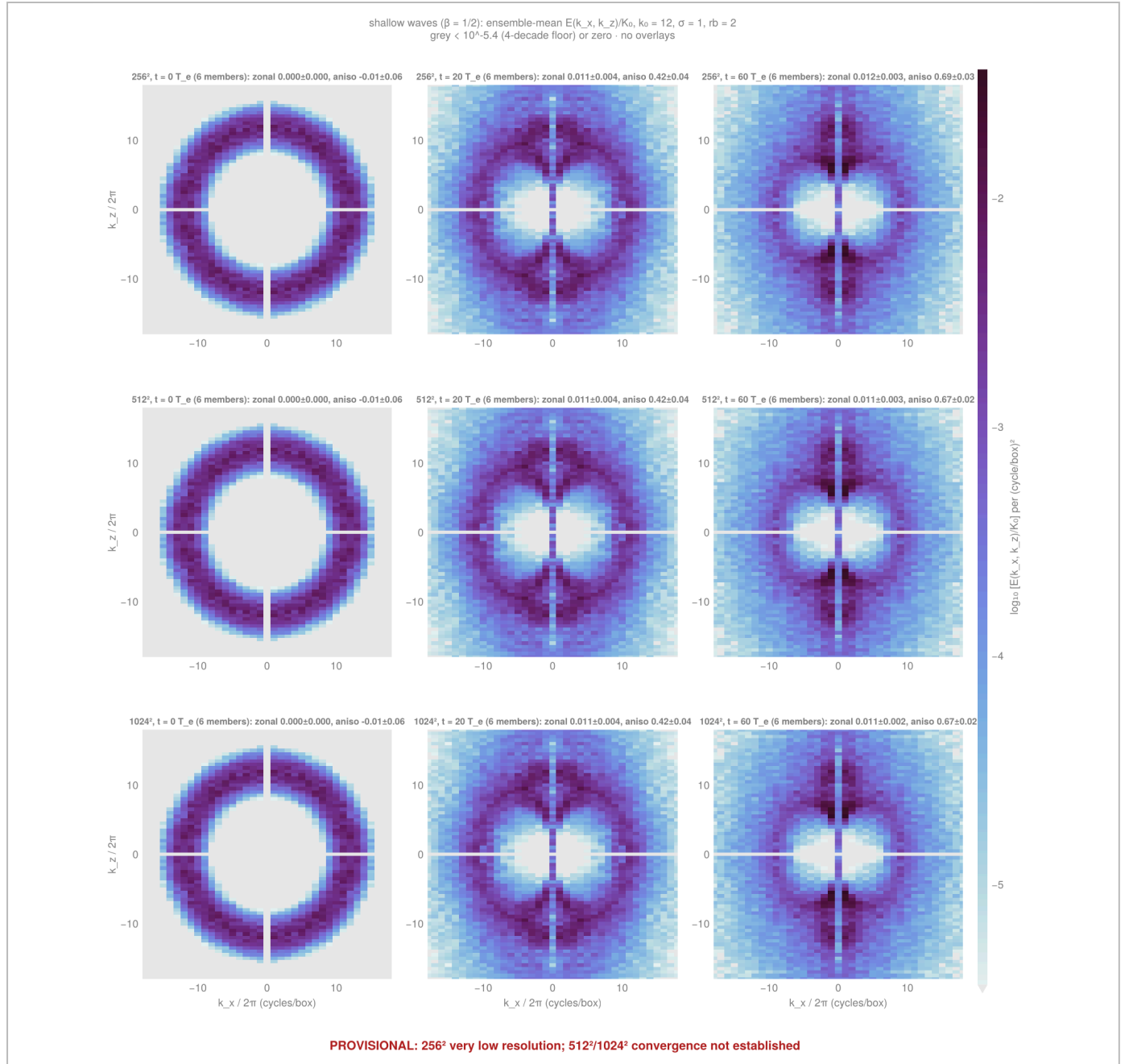


PROVISIONAL — convergence not established (1024²). VM93 Fig. 5 / Vallis Fig. 12.3 analogue (1024²). Six-seed ensemble mean $E(k_x, k_z)/K_0$ for shallow waves with VM93-inspired parameters ($k_0 = 12$, $\sigma = 1$, $rb = 2$) at 0, 20 and 60 T_e (VM93's panels sit at ≈ 0 , 17 and 50 turnovers). The energy leaves the ring, develops depleted regions near the k_x axis and concentrates in a narrow hourglass along k_z by 60 T_e . No Rhines or other estimated boundary is drawn. At 60 T_e : exact-zonal fraction 0.0110 ± 0.0024 , anisotropy 0.668 ± 0.024 , energy centroid 10.02 cycles (mean $\pm$ seed standard deviation) — the exact-zonal fraction is small, as VM93 noted for the wave regime. Heatmap: spectral density per (cycle/box)², normalized by the initial energy K_0 (Σ density $\times$ cell area = $K(t)/K_0$; checked to $1e-8$ at every panel time), one cell per lattice mode ($k_x/2\pi$ integer, $k_z/2\pi = n/2$ for the wall sine basis), $\log_{10}$ colour on a perceptually uniform sequential map with limits shared across the panels and families being compared; cells more than four decades below the shared maximum, or exactly zero, are grey. No interpolation, no overlays. *Provenance*: 1024² members ring_k12_w1_rb2_seed{s}_N1024 (all COMPLETED to 60 T_e ; IC coefficients identical to the 256² and 512² members to $\leq 4 \times 10^{-16}$ after one normalization factor). Lower grids only in the three-grid comparison below. Figure main1024_ensemble_spectra_shallow_heatmap.png.

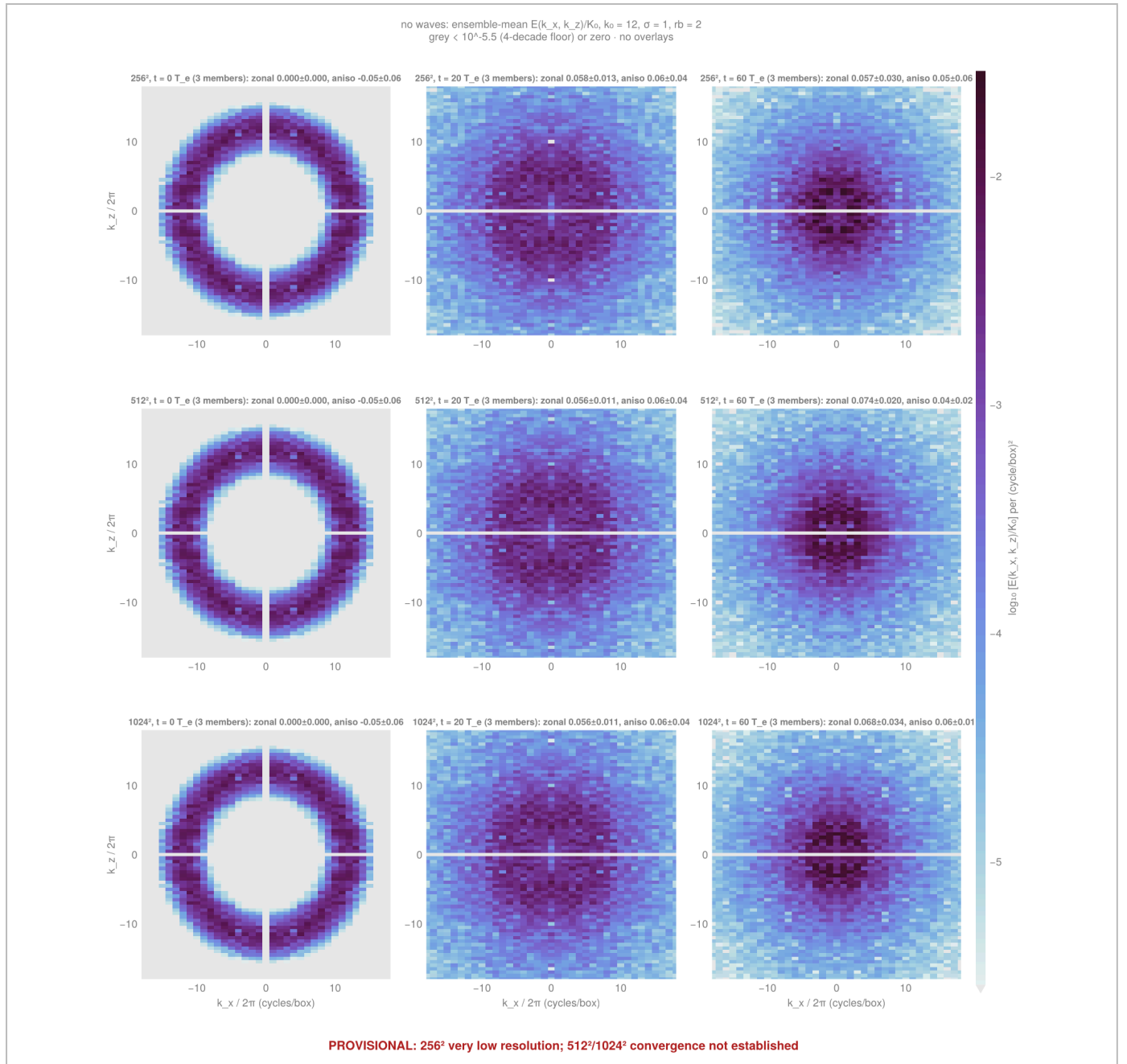


PROVISIONAL — convergence not established (1024²). Wave-free ensemble mean (1024², three seeds, same ring): isotropic collapse toward small K with no preferred direction; at 60 T_e the centroid is 5.44 cycles and the exact-zonal fraction 0.068 ± 0.034 , from the condensate's random projection onto the x -mean. Heatmap: spectral density per (cycle/box)², normalized by the initial energy K_0 (Σ density $\times$ cell area = $K(t)/K_0$; checked to $1e-8$ at every panel time), one cell per lattice mode ($k_x/2\pi$ integer, $k_z/2\pi = n/2$ for the wall sine basis), $\log_{10}$ colour on a perceptually uniform sequential map with limits shared across the panels and families being compared; cells more than four decades below the shared maximum, or exactly zero, are grey. No interpolation, no overlays. *Provenance*: 1024² members ring_k12_w1_rb2_seed{s}_N1024 (all COMPLETED to 60 T_e ; IC coefficients identical to the 256² and 512² members to $\leq 4 \times 10^{-16}$ after one normalization factor). Lower grids only in the three-grid comparison below. Figure main1024_ensemble_spectra_none_heatmap.png.

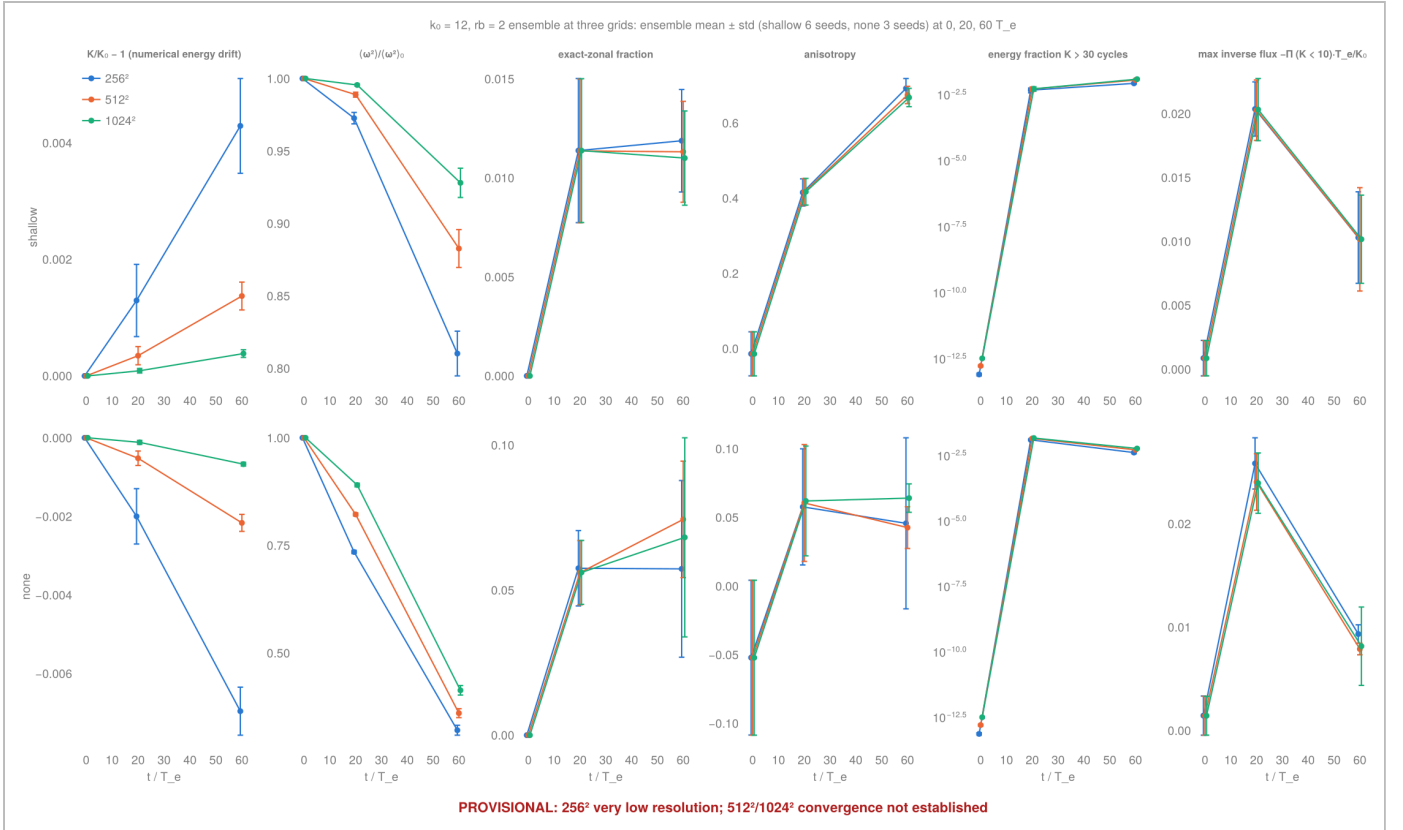
$k_0 = 12$, $rb = 2$ ensemble at 256^2 , 512^2 and 1024^2



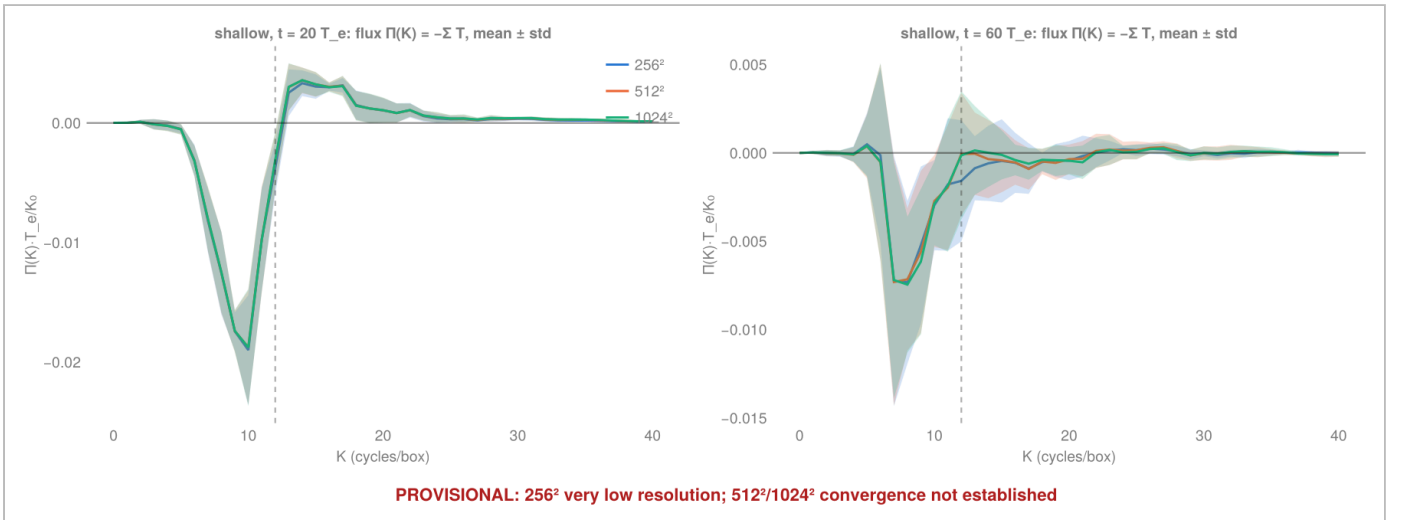
PROVISIONAL: 256² very low resolution; 512²/1024² convergence not established. $k_0 = 12$, $rb = 2$ ensemble at three grids — shallow waves. Ensemble-mean $E(k_x, k_z)/K_0$ over six seeds at 0, 20 and 60 T_e (columns) for 256², 512² and 1024² (rows). Same continuous Fourier coefficients for each seed at every grid (IC preflight passed before the ensemble ran: coefficient differences $\leq 4.1 \times 10^{-16}$ after one uniform normalization factor of 0.9974–0.9981); all 18 final cases validated complete at 60 T_e. One log₁₀ colour scale shared by all nine panels (perceptually uniform, grey below a four-decade floor or zero), one cell per mode, no overlays. Panel titles give the exact-zonal fraction and anisotropy as mean $\pm$ seed standard deviation. At 60 T_e the anisotropy is 0.692 / 0.675 / 0.668 (seed std ≈ 0.024) and the exact-zonal fraction 0.0119 / 0.0113 / 0.0110 (≈ 0.0025); the energy centroid shifts 9.78 $\rightarrow$ 9.96 $\rightarrow$ 10.02 cycles (seed std 0.15). Descriptive comparison, not a statistical test. Figure ens12_spectra_shallow_heatmap.png. See REPORT_resolution.md §6.



PROVISIONAL: 256² very low resolution; 512²/1024² convergence not established. Same comparison without waves (three seeds), same conventions; colour scale shared within this figure. Isotropic collapse toward small K at all grids; at $60 T_e$ the centroid is $5.40 / 5.37 / 5.44$ cycles and the exact-zonal fraction $0.057 \pm 0.030 / 0.074 \pm 0.020 / 0.068 \pm 0.034$. Figure ens12_spectra_none_heatmap.png. See REPORT_resolution.md §6.



PROVISIONAL: 256² very low resolution; 512²/1024² convergence not established. Selected statistics at 0, 20 and 60 T_e (mean $\pm$ seed standard deviation; shallow top, wave-free bottom): energy drift $K/K_0 - 1$, enstrophy, exact-zonal fraction, anisotropy, energy fraction at $K > 30$ cycles and the member-peak inverse flux. For shallow waves the anisotropy, exact-zonal fraction and member-peak inverse flux differ between grids by less than one seed standard deviation, with a smaller change from 512² to 1024² than from 256² to 512²; small-scale quantities are resolution dependent (enstrophy retained at 60 T_e 0.81 / 0.88 / 0.93; energy drift $+4.3 \times 10^{-3}$ / $+1.4 \times 10^{-3}$ / $+3.8 \times 10^{-4}$, i.e. better energy conservation, not proof of converged dynamics). The seed standard deviation describes member spread, not the uncertainty of the paired resolution difference. Figure ens12_statistics.png. See REPORT_resolution.md §6.



PROVISIONAL: 256² very low resolution; 512²/1024² convergence not established. Spectral energy flux $\Pi(K) = -\sum_{K' \leq K} T(K')$ for shallow waves, ensemble mean $\pm$ seed standard deviation at 20 and 60 T_e (dashed: ring $K = 12$). The three grids overlap within the spread. The member-peak values quoted elsewhere (mean over members of each member's max of $-\Pi$ for $K < 10$, $\approx 1.0 \times 10^{-2}$ after normalization by T_e/K_0 at 60 T_e , the same definition and value as the overlay audit's ϵ) exceed the peak of the ensemble-mean curve because member peaks lie at different K ; the audit's "no plateau" refers to the band average of $-\Pi$ over $2 \leq K \leq 10$, a different statistic. Figure ens12_flux.png. See REPORT_resolution.md §6.

Controls, resolution and seed dependence

CHECK	A · SURFACE STRESS	B · VALLIS-2D
Controls	Wave-free runs from the same Lagrangian field at $r = 0, 0.1, 1$: bands below the fluctuation rms at $r = 0$ ($K_{\text{jet}}/K_{\text{rel}} \leq 0.02$); roll anisotropy 1.17–1.24 at $r = 0.1, 1$ — the same as the wave cases at $La \geq 0.5$. Wave-only quiescent state exactly at rest.	Wave-free member from the identical field: Fig. 11.8 vortex decay, anisotropy ≈ 0 , remaining enstrophy 14% versus 39–42% with waves (86% lost versus 58–61%; the remaining fractions differ by about threefold, not the losses). Doubly periodic β -plane reference matches the wall case. Quiescent wave-only state exactly at rest.
Resolution	256 ³ at $r = 0, 0.03, 0.1$: surface jet $-3.21\text{e-}3$ vs $-3.24\text{e-}3$ and interior amplitude $8.4\text{e-}4$ vs $7\text{e-}4$ at $r = 0$ ($K_{\text{jet}}/K_{\text{rel}}$ 30% lower); $r = 0.1$ matches 128 ³ within 10% in u'_{rms} , u' , surface mean and K ; onset time at $r = 0.1$ shifts $82 \rightarrow 156$ while the onset ratio stays 1.02–1.06. Two grids agree on the ratio; this is consistency between two resolutions, not demonstrated convergence.	512 ² from the same coefficients: K_{jet}/K 0.050 vs 0.050 (shallow), 0.384 vs 0.400 (deep), anisotropy 0.715 vs 0.696, identical $U_L(z)$ up to small-scale detail, 6% more enstrophy at 200 T_e ; deep energy drift $+0.6\% \rightarrow +0.1\%$ over 200 T_e (decreases with resolution; not shown to vanish). Agreement between two grids; a third would be needed to claim convergence.
Seeds	Seed 2 at $r = 0, 0.03, 1$ and control 0: surface jet $-3.5\text{e-}3$ vs $-3.2\text{e-}3$, interior $8.8\text{e-}4$ vs $7\text{e-}4$ (10–20%); same transition behaviour at $r = 0.03$. One seed only at $r = 0.01, 0.1, 0.3$.	Shallow seed 2 matches seed 1 (0.043 vs ≈ 0.04); six-seed ensemble spread ± 0.003 at $k_0 = 12$; deep x-mean jet is realisation-sensitive (0.074 vs ≈ 0.2 at 100 T_e).
Time stepping	CFL 0.5 wizard, max Δt 0.1; restart bitwise identical at fixed Δt .	$\Delta t = 0.25 \dots 4$ converge to $< 1\text{e-}4$ (end-time matched); production $\Delta t = 1$ (0.79 for deep, wave limit); restart bitwise identical.

Resource use

CAMPAIGN	JOBS	BILLED GPU-H	NOTES
A · surface stress	31	10.83	128 ³ sweep + extensions to $t = 1000$ (≈ 7 GPU-h), three 256 ³ cases (1.5), seed 2 (1.0), medium family (0.4), validation (0.9), analysis and animations (0.4)
B · Vallis-2D	12	2.11	validation (0.10), pilots incl. the env-truncation rerun (1.13), 512 ² (0.19), β -plane (0.07), sensitivity (0.24), ensemble (0.23), figure passes (0.15); a 256 ² member costs 1–3 min once output I/O is sparse
total	43	12.94	ceiling 60 GPU-h (Vallis sub-budget 8); at most 4 concurrent single-GPU jobs; partition default memory. Job-level ledger: <code>job_ledger.md</code> .

Caveats

Deep-wave discrete energy drift (Vallis-2D). In the continuum the Stokes terms do no work on u^L , and the shallow and wave-free members conserve energy to ± 0.03 – 0.6% . The deep member gains $+0.6\%$ over $200 T_e$ at 256^2 and $+0.1\%$ at 512^2 : the staggered-grid interpolation of $\partial_z u^s(w \hat{x} - u \hat{z})$ is not exactly energy-neutral when $\partial_z u^s$ varies on the grid scale (e-folding depth $1/8 = 32$ cells at 256^2). The drift is a discretisation residual that decreases with resolution ($+0.6\% \rightarrow +0.1\%$ between the two grids over $200 T_e$; it has not been shown to vanish) and does not alter the qualitative picture, but deep-wave energy budgets at 256^2 should not be read to better than 1% .

Surface-stress thresholds are observations in a sampled setup. See §A: single seeds at most ratios, two grids, strongly dissipative 128³ sweep, weak jets relative to WC25, box-confined bands, velocity-dependent implicit dissipation at $La \geq 0.5$, finite durations. No universal transition is asserted and no claim is made that weak stresses eventually produce rolls; $u^*/u'_{rms} \gtrsim 0.7$ – 1 with $La \lesssim 0.3$ is where rolls were seen in these runs.

- **Not quantitative reproductions.** A: $u'_{rms,0}$ and k_e are pilot choices and WC25's spun-up IC was not reproduced. B: VM93's conventions are inferred ($E = u'_{rms,0}{}^2/2$, $Z = \langle \zeta^2 \rangle$, 2π box) and Vallis' Fig. 11.8 IC is known only as "a few modes around wavenumber 9"; the comparison is the dumbbell shape and zonal elongation, not amplitudes.
- **Finite time.** B's shallow case at $rb = 1$ has not formed finite-amplitude exact-zonal jets by $200 T_e$; the Rhines wavenumber coincides with the ring by construction. A's $r = 1$ to $t \approx 1000$ and the 256³ cases stop at $t = 400$ – 1000 .
- **Numerics.** WENO(9) Z-weights use $\epsilon = 1e-8$; at $u'_{rms,0} \sim 1e-4$ (B) the scheme is effectively a linear 9th-order upwind scheme, tested by the $\lambda = 200$ member (same phenomenology, 1% more dissipation). The pseudo-spectral reference uses ∇^8 hyperviscosity, so its dissipation differs by construction.
- **Walls and symmetry (B).** The sine basis makes the k_z lattice spacing π (mode counts not isotropic) and removes $k_x = 0$ modes; free-slip walls admit boundary-trapped x-mean flow (deep case, $z \approx 0.9$).

- **Sign convention.** WC25 prints $\partial z u^s = \pm z/2$ in different places; both campaigns use $+z/2$, which fixes the Rossby phase direction ($-x$) and nothing else.
- **Operational.** A's first validation attempt and B's first two failed on test-side bugs (fixed, cost 0.8 GPU-h in total); B's first pilot lost its 512² members to whitespace truncation in the submission tool's environment passing (fixed on both sides); all rerun.

Editorial corrections to the source reports

Both campaign reports preserve the original results, with updated velocity notation and dated, bracketed correction notes (REPORT_surface_stress.md, REPORT_vallis2d.md) for traceability. Following an independent review of the surface-stress report, this synthesis departs from the source wording in the following places; these changes correct definitions and interpretation without changing simulation data.

- **Langmuir number at zero stress.** The source lists $La = \infty$ for $u^*/u'_{rms,0} = 0$. With $La = \sqrt{(u^*/u^s(1))}$, zero stress with finite waves gives $La = 0$. For the wave-free controls we label La not applicable: the formula is singular for positive stress and indeterminate at zero stress. The case table above uses 0 and "undefined" accordingly.
- **"Every finite stress eventually crosses the first threshold."** Dropped. The finite-time runs show $u^*/u'_{rms}(t)$ increasing in every stressed case, but the stress also sustains fluctuations (plateauing enstrophy), so an eventual crossing for weak stress is not established.
- **"Regime rule".** Rephrased as the pattern observed in this sampled sweep (one seed at most ratios, two grids at three ratios), not as necessary or sufficient conditions for rolls.
- **"Resolution test of the rule" / "256³ confirms".** Rephrased as agreement between two grids; convergence is not demonstrated by two resolutions.
- **"Exceed CFL-safe velocities".** The high-stress cases are integrated at CFL 0.5 by the adaptive step; the caveat is that $O(1)$ mean advection makes the implicit WENO dissipation velocity-dependent (Galilean-invariance caveat for the LES), which is distinct from a CFL stability problem.
- **"The Lagrangian mean stays nearly shear-free".** Qualified: near-shear-free relative to the Stokes profile ($U_E \approx -u^s$ persists), apart from the surface jet and interior bands that are the subject of the campaign.
- **Final times and onset measures.** The deep $r = 1$ extension was still finishing at $t \approx 1000$ when the report was written ($t = 400$ is the completed common end time). *Corrected 2026-10-03 (classifier audit):* an earlier version of this item attributed the 256³ $r = 0.03$ onset ($t \approx 270$) to a layer-restricted measure and called the timing uncertain because it disagreed with the raw volume ratio (≈ 1.2 at $t = 400$). Both statements were wrong. The $t \approx 270$ onset comes from the depth-mean fluctuation ratio R_{fl} defined below; the raw volume ratio is a different quantity that includes the mean shear and is not a roll classifier. With consistent classifiers the two grids agree: R_{fl} onset $t = 277$ (128³) and 270 (256³), layer-restricted R_{layer} onset $t = 178$ and 167. The legacy 128³ and 256³ initial conditions are different realizations, so these are realization-plus-resolution comparisons (see REPORT_resolution.md).
- **Roll-ratio definitions and the primed raw ratio (added 2026-10-03).** Standard curl signs: $\omega_x = \partial_y w - \partial_z v$ (along-wave axis), $\omega_y = \partial_z u - \partial_x w$ (cross-wave axis). $\mathbf{R}_{raw} = \langle \omega_x^2 \rangle_V / \langle \omega_y^2 \rangle_V$: ratio of volume means of the *total* squared components (scalar statistics), which include the mean-shear terms $(\partial_z V)^2$ and $(\partial_z U)^2$; the wind layer's $(\partial_z U)^2$ keeps it near 1 even when rolls are present. $\mathbf{R}_{fl} = \langle \omega'_x{}^2 \rangle / \langle \omega'_y{}^2 \rangle$ (the roll classifier for all onset times and the ratio plotted in the anisotropy and regime figures): horizontal means of ω_x^2 , ω_y^2 on the $N + 1$ z -faces, minus $(\partial_z V)^2$ and $(\partial_z U)^2$ from the mean profiles (an exact removal of the mean-shear part), averaged with equal weights over all faces, then divided. $\mathbf{R}_{layer}$: the same with the depth mean restricted to faces with $z > 0.75$. Onset: first $t > 5$ with the ratio > 1.3 . The source report's sentence "a slightly earlier roll onset ($\langle \omega_x^2 \rangle / \langle \omega_y^2 \rangle \approx 1.2$ at $t = 400$ in the raw measure)" for 256³ $r = 0.03$ primes a raw quantity: the 1.2 is R_{raw} (1.175 at

256³ vs 1.003 at 128³); $R_{fl}(400)$ is 1.658 (256³) and 1.757 (128³), and the corresponding onset is slightly earlier at 256³ (270 versus 277), rather than evidence of a disagreement between classifiers. Figure labels were checked: every primed ratio in the figures is R_{fl} .

- **Resolution follow-up (interim, added 2026-10-03).** Same-field 128³/256³/512³ initial conditions, a 1024² two-dimensional baseline and an accelerating-frame check are reported in REPORT_resolution.md. The 512³ runs and the 256³ $r = 0.3$ pair are still pending; nothing in this document is revised on their basis yet.

The Vallis-2D report is by the same author as this synthesis; following a second independent review it is corrected here as follows:

- **Deep energy drift.** The source says the Stokes-term residual "vanishes with resolution"; the data show +0.6% (256²) and +0.1% (512²) over 200 T_e , i.e. a decrease between two grids, and this synthesis says so.
- **Ring width.** The source infers a " ≈ 1 -mode" VM93 ring from $Z/E = 2K^2$; that ratio is consistent with a narrow ring but does not determine the width, so $\sigma = 1$ mode is described here as a chosen value with $\sigma = 0.5$ and 2 as sensitivity tests.
- **Deep seed comparison.** Compared at equal times: seed 2 gives $K_{jet}/K = 0.074$ at 100 T_e against ≈ 0.2 for seed 1 at 100 T_e (not against the 0.39 that seed 1 reaches at 200 T_e).
- **No spectral-boundary overlay.** The main spectra draw no Rhines or Vallis Eq. 12.14 curve. A global velocity-based estimate is not the observed hole boundary, and Eq. 12.14 needs a defensible inverse transfer rate that the sampled fluxes do not provide (REPORT_overlay_audit.md).
- **"Resolution-converged at 256²"** is softened to agreement between 256² and 512²; "reproduced" means a source-matched qualitative comparison, not an exact reproduction.

Paths, commits and data

	A · SURFACE STRESS	B · VALLIS-2D
Code	/u/glwagner/nwt_wc25_surface_stress, branch glw/wc25-surface-stress (HEAD 8c5bc97): experiments/wc25_surface_stress, analysis/wc25_surface_stress, batch/wc25_*.batch	/u/glwagner/nwt_wc25_vallis2d, branch glw/wc25-vallis2d (HEAD f21278d + this document): experiments/wc25_vallis2d, analysis/wc25_vallis2d, batch/vallis_*.batch
Data	/work/hdd/bhcr/glwagner/wc25_surface_stress_2026-10-02/<waves>_r<ratio>_seed<s>_N<N>/	/work/hdd/bhcr/glwagner/wc25_vallis2d_2026-10-02/<waves>_ring_k<k0>_w<sigma>_rb<rb>_seed<s>_N<N>/, betaplane_*/, initial_conditions/
Figures	figures/wc25_surface_stress/{pilot1,sweep1,sweep2,final,final256,final_medium,robust}_*.png, jets_vs_rolls_sections.mp4	figures/wc25_vallis2d/{pilot,final}_*.png, final_metrics.txt
Reports	experiments/wc25_surface_stress/REPORT.md (bundled as REPORT_surface_stress.md)	figures/wc25_vallis2d/REPORT.md (bundled as REPORT_vallis2d.md)
Status / ledger	/u/glwagner/wc25_surface_stress_status.md, wc25_surface_stress_coordination/ledger/case_table.md	/u/glwagner/wc25_vallis2d_status.md, wc25_vallis2d_coordination/{submit_requests,submit_responses}

PNG figures are ignored by the repository's .gitignore and live on disk only (the surface-stress campaign force-added its final set). This document references them by relative path from figures/wc25_results/; the review bundle preserves that layout. Source movies and the pre-existing packet-wake gallery content are served from the published glw/anti-stokes branch and are not duplicated in the bundle.

References

- Wagner & Constantinou (2025), *Phenomenology of decaying turbulence beneath surface waves*, JFM 1020, A51. doi:10.1017/jfm.2025.10649.
- Vallis & Maltrud (1993), *Generation of Mean Flows and Jets on a Beta Plane and over Topography*, JPO 23, 1346–1362. Author PDF.
- Vallis (2017), *Atmospheric and Oceanic Fluid Dynamics*, second edition, Cambridge University Press, Figs 11.8, 12.3 and 12.4. Related author teaching code.